

Product ID	AT2222
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 179: AT2222 - technical data (cont.)

12.2.4 Additional technical data

Product ID	AT2222
Temperature inputs - resistance measurement	
Sensor standard	IEC/EN 60751
Temperature sensor resolution PT100 PT1000	1 LSB = 0.1 °C 1 LSB = 0.1 °C
Resistance measurement resolution G = 1 G = 2	0.1 Ω 0.05 Ω
Temperature sensor standardization PT100 PT1000	-200.0 °C to +850.0 °C -200.0 °C to +850.0 °C
Standardized value range for resistance measurement G = 1 G = 2	0.1 Ω to 4500 Ω 0.05 Ω to 2250 Ω
Temperature measurement monitoring Open inputs Wire break Below lower range limit Above upper range limit General error	\$7FFF \$7FFF \$8001 \$7FFF \$8000
Resistance measurement monitoring Open inputs Wire break Above upper range limit General error	\$FFFF \$FFFF \$FFFF \$FFFF
Conversion method	Sigma Delta
Linearization method	Internal
Reference	4530 Ω ±0.1 %
Measuring current	250 µA ±1.25 %
Input filter	Low pass 1st order / cut-off frequency 500 Hz
Maximum gain drift	<0.004 % per °C, based on the current resistance measurement
Maximum offset drift	<0.00015 % per °C, based on the entire resistance measurement range

Table 180: AT2222 - additional technical data

Product ID	AT2222
Common-mode rejection DC 50 Hz	>95 dB >80 dB
Synchronized zone	>0.7 V
Cross-talk between channels	>93 dB
Non-linearity	<0.0010 %, based on the entire resistance range
Isolation voltage betw. channel and bus	500 V _{eff}
General information	
B&R ID code	\$1BA6

Table 180: AT2222 - additional technical data (cont.)

12.2.5 Status LEDs

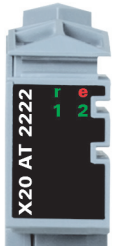
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
			Single flash	Warning/error for an I/O channel. Overflow or underflow of the analog inputs.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 2	Green	Off	The input is switched off
			Blinking	Overflow, underflow or broken connection
			On	The analog/digital converter is running

Table 181: AT2222 - status indicators

12.2.6 Pin assignments

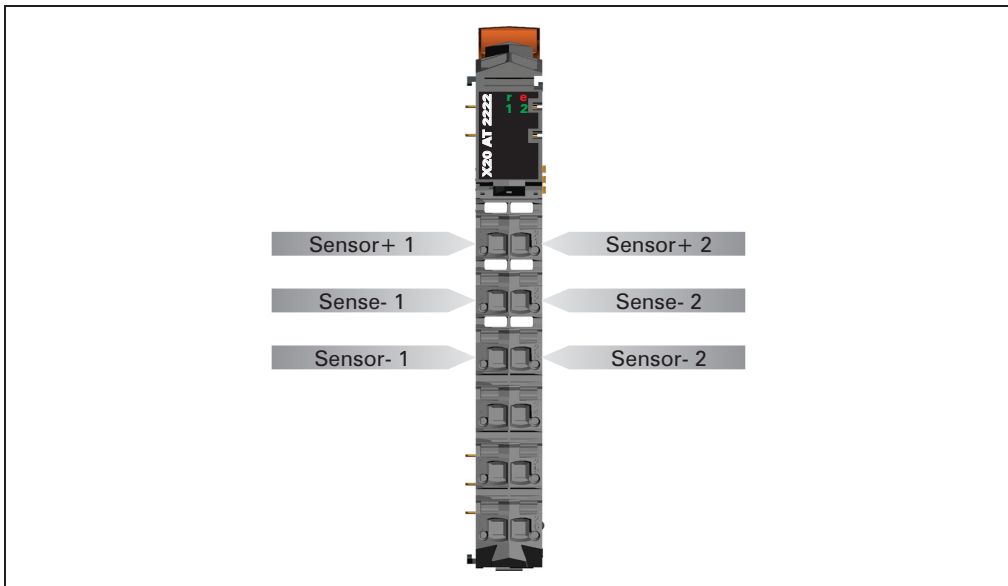


Figure 134: AT2222 - pin assignments

12.2.7 Connection example

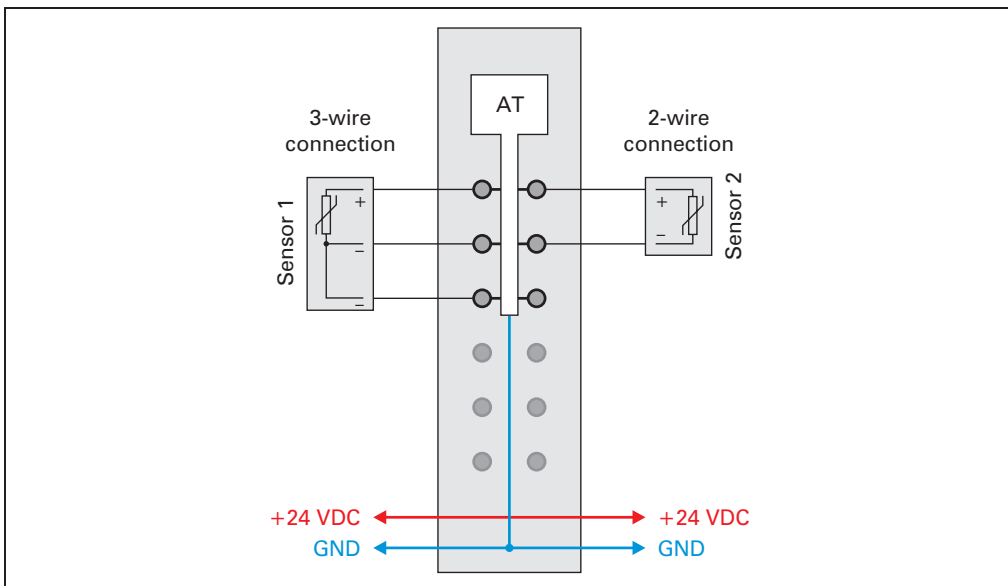


Figure 135: AT2222 - connection example

12.2.8 Input circuit diagram

2-wire connection

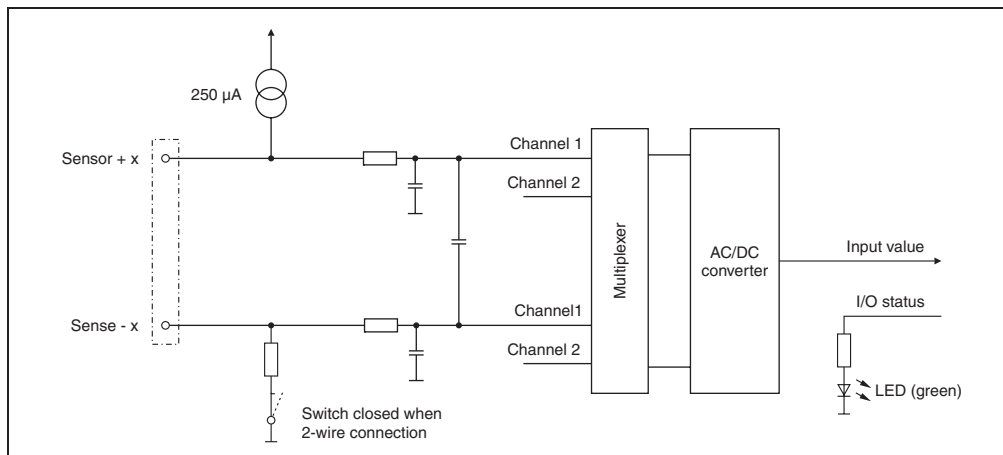


Figure 136: AT2222 - input circuit diagram - 2-wire connection

3-wire connection

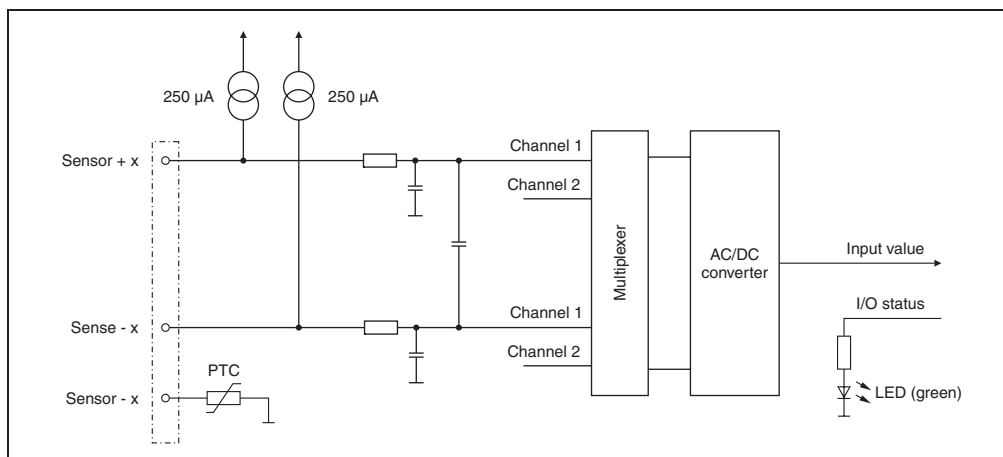


Figure 137: AT2222 - input circuit diagram - 3-wire connection

12.2.9 Analog inputs

The converted analog values are output by the module in the registers. Different resistance or temperature measurements will result in different value ranges and data types.

12.2.10 Timing setting

The timing setting for data acquisition is made using the converter hardware. All switched on inputs are converted during each conversion cycle.

12.2.11 Conversion time

The conversion time for the channels depends on their use. For the formulas listed in the table, 'n' corresponds to the number of channels that are switched on.

Channel uses	Conversion time
1 channel	$1 \cdot \text{FilterTime}$
n channels with the same sensor type	$n \cdot 20\text{ms} + \text{FilterTime}$
n channels with different sensor types	$n \cdot 20\text{ms} + 2 \cdot \text{FilterTime}$

12.2.12 Reduced refresh time

If an input is not necessary, it can be switched off, thereby reducing the refresh time. Inputs can also be only temporarily switched off.

The time saved is equal to:

$$\text{TimeSaved} = 2 \cdot 20\text{ms} + \text{FilterTime}$$

The filter time is the conversion time for the remaining channels.

Examples

The inputs are filtered using a 60 Hz filter.

	Example 1	Example 2
Switched on inputs	1	1 - 2
Conversion time	16.7 ms	73.4 ms

12.2.13 Input filter

The filter time for all analog inputs is defined using the input filter parameter.

Value	Filter	Filter time	Digital converter resolution
0	15 Hz	66.7 ms	16-bit
1	25 Hz	40 ms	16-bit
2	30 Hz	33.3 ms	16-bit
3	50 Hz	20 ms	16-bit
4	60 Hz	16.7 ms	16-bit
5	100 Hz	10 ms	16-bit
6	500 Hz	2 ms	16-bit
7	1000 Hz	1 ms	16-bit

12.2.14 Sensor type and channel deactivation

The module is designed for temperature and resistance measurement. The sensor type must be specified because of the different adjustment values for temperature and resistance.

The default setting for all channels is ON. To save time, individual channels can be switched off (see section 12.2.12 "Reduced refresh time" on page 325).

Code	Input signal
0	Reserved
1	Reserved
2	Sensor type PT100
3	Sensor type PT1000
4	Reserved
5	Resistance measurement 0.1 Ω to 4500 Ω
6	Resistance measurement 0.05 Ω to 2250 Ω
7	Channel turned off

12.2.15 Input status

The module's inputs are monitored. A change in the monitoring status generates an error message.

Code	Channel x
0	No error
1	Below lower limit value
2	Above upper limit value
3	Wire break

12.2.16 "StatusInput01" register

Bit	Description
0 - 1	Channel 1: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break
2 - 3	Channel 2: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break
4 - 7	0

In addition to the status info, the error type also sets the analog value to the following values:

Error type	Temperature measurement Digital value for error	Resistance measurement Digital value for error
Wire break	+32767 (\$7FFF)	65535 (\$FFFF)
Above upper limit value	+32767 (\$7FFF)	65535 (\$FFFF)
Below lower limit value	-32767 (\$8001)	0 (\$0000)
Invalid value	-32768 (\$8000)	65535 (\$FFFF)

12.2.17 IOcyclicCounter

The cyclic counter increases after all input data have been updated.

12.2.18 B&R ID code

Code for module identification (\$1BA6).

12.2.19 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
In each operating mode	≥100 µs

12.2.20 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
1 input	Corresponds to the filter time
2 inputs	$2 \cdot (20\text{ms} + \text{FilterTime})$

12.3 AT4222

12.3.1 General information

The AT4222 module is equipped with four inputs for PT100/PT1000 resistance temperature measurement.

- 4 inputs for resistance temperature measurement
- For PT100 and PT1000
- Sensor type can be set for each channel
- Direct resistance measurement as well
- 2 or 3-wire connection can be configured for each module
- Filter time can be configured

12.3.2 Order data

Model number	Short description	Image
	Temperature module	
X20AT4222	X20 temperature input module, 4 resistance measurement inputs, PT100, PT1000, resolution 0.1 K, 3-wire connections	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 182: AT4222 - order data

12.3.3 Technical data

Product ID	AT4222
Short description	
I/O module	4 inputs for PT100 or PT1000 resistance temperature measurement
Temperature inputs - resistance measurement	
Input	Resistance measurement with constant current supply for 2 or 3-wire connection
Digital converter resolution	16-bit
Filter time	Configurable between 1 ms and 66.7 ms
Conversion time 1 channel 2 - 4 channels	20 ms at 50 Hz filter 40 ms per channel with 50 Hz filter
Output format	INT or UINT for resistance measurement
Maximum error at 25 °C Gain Offset	<0.037 %, based on the current resistance value <0.0015 %, based on the entire resistance range
Sensor PT100 PT1000	Can be set per channel -200 °C to +850 °C -200 °C to +850 °C
Resistance measurement range	0.1 Ω to 4500 Ω / 0.05 Ω to 2250 Ω
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Inputs	Yes, with status LED and software status Yes, with status LED and software status
Electrical isolation Channel - Bus Channel - Channel	Yes No
Power consumption Bus I/O internal	Typ. 0.01 W Typ. 1.1 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20

Table 183: AT4222 - technical data

Product ID	AT4222
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 183: AT4222 - technical data (cont.)

12.3.4 Additional technical data

Product ID	AT4222
Temperature inputs - resistance measurement	
Sensor standard	IEC/EN 60751
Temperature sensor resolution PT100 PT1000	1 LSB = 0.1 °C 1 LSB = 0.1 °C
Resistance measurement resolution G = 1 G = 2	0.1 Ω 0.05 Ω
Temperature sensor standardization PT100 PT1000	-200.0 °C to +850.0 °C -200.0 °C to +850.0 °C
Standardized value range for resistance measurement G = 1 G = 2	0.1 Ω to 4500 Ω 0.05 Ω to 2250 Ω
Temperature measurement monitoring Open inputs Wire break Below lower range limit Above upper range limit General error	\$7FFF \$7FFF \$8001 \$7FFF \$8000
Resistance measurement monitoring Open inputs Wire break Above upper range limit General error	\$FFFF \$FFFF \$FFFF \$FFFF
Conversion method	Sigma Delta
Linearization method	Internal
Reference	4530 Ω ±0.1 %
Measuring current	250 µA ±1.25 %
Input filter	Low pass 1st order / cut-off frequency 500 Hz
Maximum gain drift	<0.004 % per °C, based on the current resistance measurement
Maximum offset drift	<0.00015 % per °C, based on the entire resistance measurement range

Table 184: AT4322 - additional technical data

Product ID	AT4222
Common-mode rejection DC 50 Hz	>95 dB >80 dB
Synchronized zone	>0.7 V
Cross-talk between channels	>93 dB
Non-linearity	<0.0010 %, based on the entire resistance range
Isolation voltage betw. channel and bus	500 V _{eff}
General information	
B&R ID code	\$1BA7

Table 184: AT4322 - additional technical data (cont.)

12.3.5 Status LEDs

Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
			Single flash	Warning/error for an I/O channel. Overflow or underflow of the analog inputs.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 4	Green	Off	The input is switched off
			Blinking	Overflow, underflow or broken connection
			On	The analog/digital converter is running

Table 185: AT4222 - status indicators

12.3.6 Pin assignments

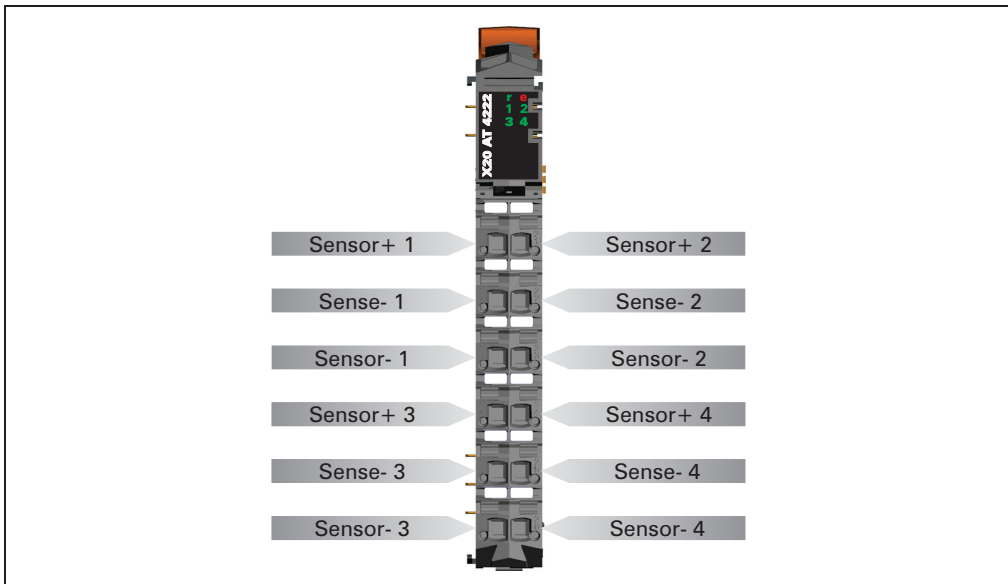


Figure 138: AT4222 - pin assignments

12.3.7 Connection example

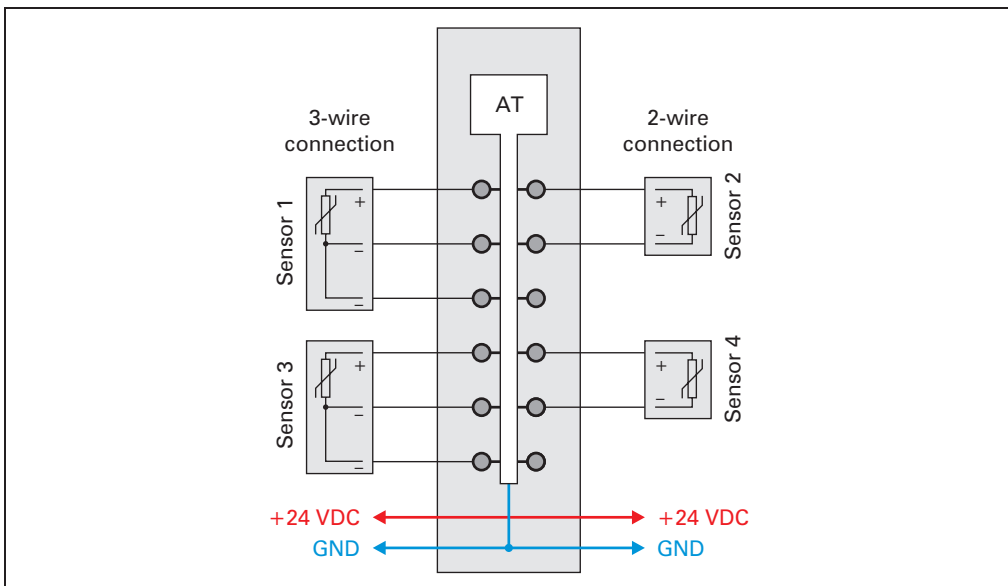


Figure 139: AT4222 - connection example

12.3.8 Input circuit diagram

2-wire connection

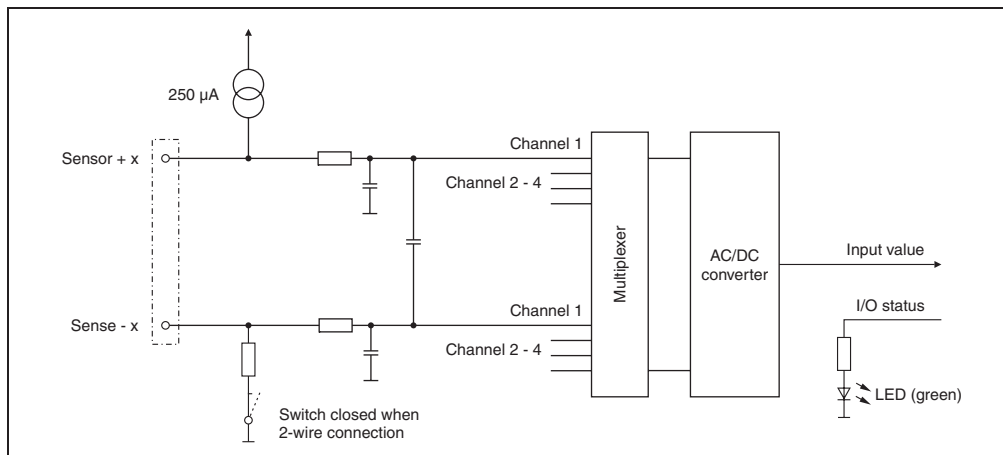


Figure 140: AT4222 - input circuit diagram - 2-wire connection

3-wire connection

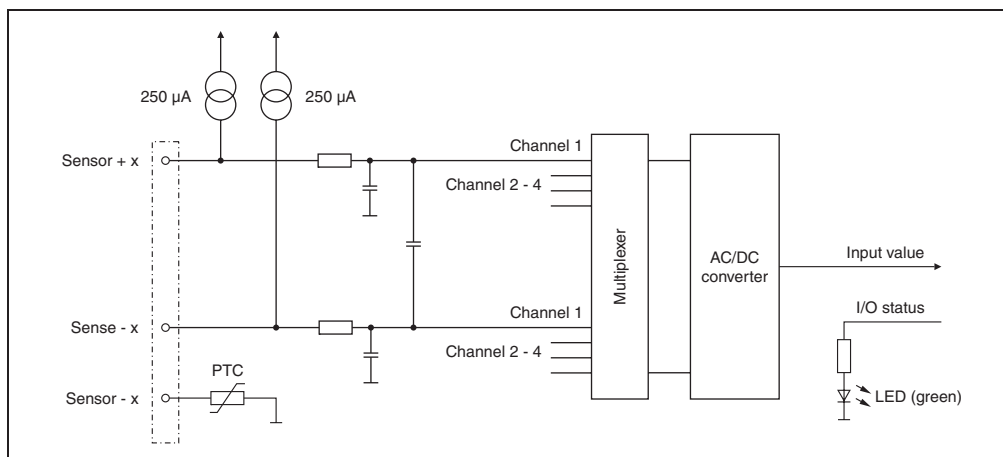


Figure 141: AT4222 - input circuit diagram - 3-wire connection

12.3.9 Analog inputs

The converted analog values are output by the module in the registers. Different resistance or temperature measurements will result in different value ranges and data types.

12.3.10 Timing setting

The timing setting for data acquisition is made using the converter hardware. All switched on inputs are converted during each conversion cycle.

12.3.11 Conversion time

The conversion time for the channels depends on their use. For the formulas listed in the table, 'n' corresponds to the number of channels that are switched on.

Channel uses	Conversion time
1 channel	$1 \cdot \text{FilterTime}$
n channels with the same sensor type	$n \cdot 20\text{ms} + \text{FilterTime}$
n channels with different sensor types	$n \cdot (20\text{ms} + 2 \cdot \text{FilterTime})$

12.3.12 Reduced refresh time

If an input is not necessary, it can be switched off, thereby reducing the refresh time. Inputs can also be only temporarily switched off.

Calculating the time saved

The amount of time saved can be calculated with the following formula. And 'n' corresponds to the number of inputs that are switched off.

$$\text{TimeSaved} = n \cdot (20\text{ms} + \text{FilterTime})$$

Examples

The inputs are filtered using a 60 Hz filter.

	Example 1	Example 2	Example 3
Switched on inputs	1	1, 3	1 - 4
Conversion time	16.7 ms	73.4 ms	146.8 ms

12.3.13 Input filter

The filter time for all analog inputs is defined using the input filter parameter.

Value	Filter	Filter time	Digital converter resolution
0	15 Hz	66.7 ms	16-bit
1	25 Hz	40 ms	16-bit
2	30 Hz	33.3 ms	16-bit
3	50 Hz	20 ms	16-bit
4	60 Hz	16.7 ms	16-bit
5	100 Hz	10 ms	16-bit
6	500 Hz	2 ms	16-bit
7	1000 Hz	1 ms	16-bit

12.3.14 Sensor type and channel deactivation

The module is designed for temperature and resistance measurement. The sensor type must be specified because of the different adjustment values for temperature and resistance.

The default setting for all channels is ON. To save time, individual channels can be switched off (see section 12.3.12 "Reduced refresh time" on page 335).

Code	Input signal
0	Reserved
1	Reserved
2	Sensor type PT100
3	Sensor type PT1000
4	Reserved
5	Resistance measurement 0.1 Ω to 4500 Ω
6	Resistance measurement 0.05 Ω to 2250 Ω
7	Channel turned off

12.3.15 Input status

The module's inputs are monitored. A change in the monitoring status generates an error message.

Code	Channel x
0	No error
1	Below lower limit value
2	Above upper limit value
3	Wire break

12.3.16 "StatusInput01" register

Bit	Description
0 - 1	Channel 1: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break
2 - 3	Channel 2: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break
4 - 5	Channel 3: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break
6 - 7	Channel 4: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break

In addition to the status info, the error type also sets the analog value to the following values:

Error type	Temperature measurement Digital value for error	Resistance measurement Digital value for error
Wire break	+32767 (\$7FFF)	65535 (\$FFFF)
Above upper limit value	+32767 (\$7FFF)	65535 (\$FFFF)
Below lower limit value	-32767 (\$8001)	0 (\$0000)
Invalid value	-32768 (\$8000)	65535 (\$FFFF)

12.3.17 IOcyclicCounter

The cyclic counter increases after all input data have been updated.

12.3.18 B&R ID code

Code for module identification (\$1BA7).

12.3.19 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
In each operating mode	$\geq 100 \mu\text{s}$

12.3.20 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
1 input	Corresponds to the filter time
n inputs	$n \cdot 20\text{ms} + \text{FilterTime}$

12.4 AT2402

12.4.1 General information

The AT2402 module is equipped with two inputs for J, K, N, and S thermocouple sensors. The module has an integrated terminal temperature compensation.

The AT2402 is designed for standard X20 6-pin terminal blocks. If needed (e.g. for logistical reasons), the 12-pin terminal block can also be used.

- 2 inputs for thermocouples
- For sensor types J, K, N, S
- Additional direct raw value measurement
- Integrated terminal temperature compensation
- Filter time can be configured

12.4.2 Order data

Model number	Short description	Image
	Temperature module	
X20AT2402	X20 temperature input module, 2 thermocouple inputs, type J,K,N,S, resolution 0.1 K	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 186: AT2402 - order data

12.4.3 Technical data

Product ID	AT2402
Short description	
I/O module	2 inputs for thermocouples
Thermocouple temperature inputs	
Input	Thermocouple
Digital converter resolution	16-bit
Filter time	Configurable between 1 ms and 66.7 ms
Conversion time	
1 channel	80.4 ms at 50 Hz filter
2 channels	120.6 ms at 50 Hz filter
Output format	INT
Basic accuracy	
Type J	$\pm 0.10\%$ at 25 °C ¹⁾
Type K	$\pm 0.11\%$ at 25 °C ¹⁾
Type N (Rev. $\geq D0$)	$\pm 0.11\%$ at 25 °C ¹⁾
Type S	$\pm 0.17\%$ at 25 °C ¹⁾
Measurement range	
Sensor temperature	
FeCuNi: Type J	-210 °C to +1200 °C
NiCrNi: Type K	-270 °C to +1372 °C
NiCrSi: Type N (Rev. $\geq D0$)	-270 °C to +1300 °C
PTRhPt: Type S	-50 °C to +1768 °C
Terminal temperature	0 °C to +85 °C
Raw value	± 65.534 mV
Terminal temperature compensation	Internal
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics	
Module run error	Yes, with status LED and software status
Inputs	Yes, with status LED and software status
Electrical isolation	
Channel - Bus	Yes
Channel - Channel	No
Power consumption	
Bus	Typ. 0.01 W
I/O internal	Typ. 0.72 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature	
Horizontal installation	0 °C to +55 °C
Vertical installation	0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical

Table 187: AT2402 - technical data

Product ID	AT2402
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 187: AT2402 - technical data (cont.)

1) Based on the measurement range, without consideration of the reference junction measurement error.

12.4.4 Additional technical data

Product ID	AT2402
Thermocouple temperature inputs	
Sensor standard	IEC/653/225/DIS
Resolution Sensor temperature Terminal temperature Raw value output with respect to gain	1 LSB = 0.1 °C 1 LSB = 0.1 °C 1 LSB = 1 µV or 2 µV Gain setting - see section 12.4.16 "Sensor type" on page 346
Standardization Type J Type K Type N (Rev. ≥D0) Type S Terminal temperature	-210.0 °C to +1200.0 °C -270.0 °C to +1372.0 °C -270.0 °C to +1300.0 °C -50.0 °C to +1768.0 °C -25.0 °C to +85.0 °C
Monitoring Open inputs Wire break Below lower range limit Above upper range limit General error	\$7FFF \$7FFF \$8001 \$7FFF \$8000
Conversion method	Sigma Delta
Linearization method	Internal
Input filter	Low pass 1st order / cut-off frequency 500 Hz
Maximum gain drift	<0.0247 % per °C, based on the current measurement value
Maximum offset drift Type J Type K Type N (Rev. ≥D0) Type S	<0.0025 % per °C, based on the entire measurement range <0.0031 % per °C, based on the entire measurement range TBD <0.0092 % per °C, based on the entire measurement range

Table 188: AT2402 - additional technical data

Product ID	AT2402
Terminal temp. compensation precision with natural convection with artificial convection	± 2 °C after 10 min ± 4 °C after 10 min
Common-mode rejection DC 50 Hz	>77 dB >77 dB
Synchronized zone	± 15 V
Cross-talk between channels	-70 dB
Non-linearity	TBD
Isolation voltage betw. channel and bus	500 V _{eff}
General information	
B&R ID code	\$1BA8

Table 188: AT2402 - additional technical data (cont.)

12.4.5 Status LEDs

Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
			Single flash	Warning/error for an I/O channel. Overflow or underflow of the analog inputs.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 2	Green	Off	The input is switched off
			Blinking	Overflow, underflow or broken connection
			On	The analog/digital converter is running

Table 189: AT2402 - status indicators

12.4.6 Pin assignments

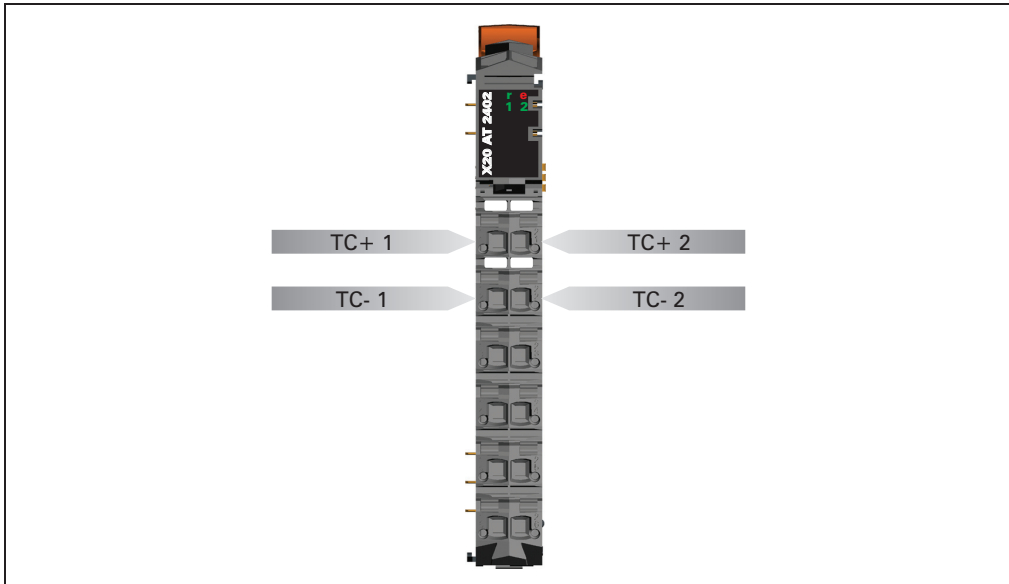


Figure 142: AT2402 - pin assignments

12.4.7 Connection example

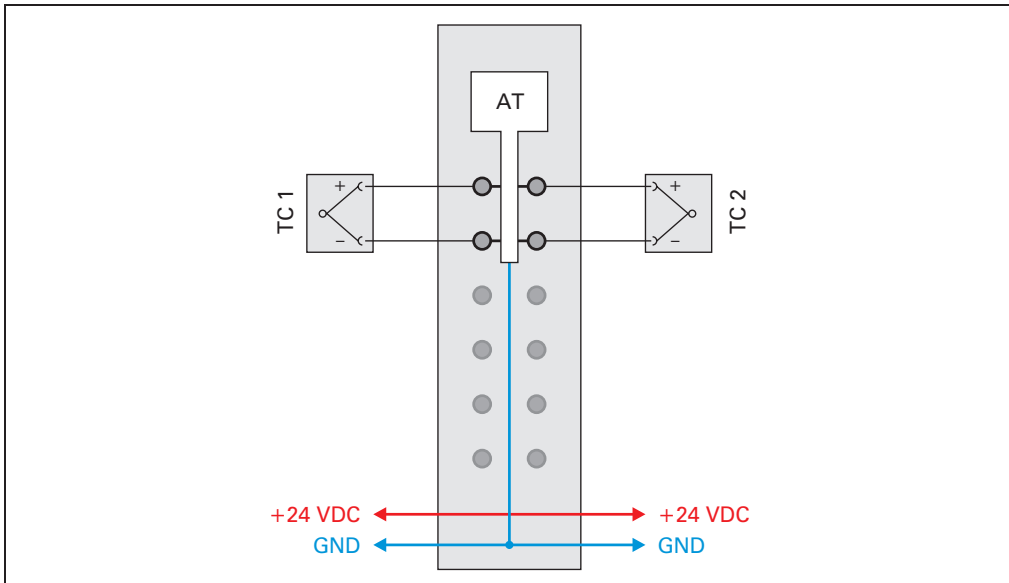


Figure 143: AT2402 - connection example

12.4.8 Input circuit diagram

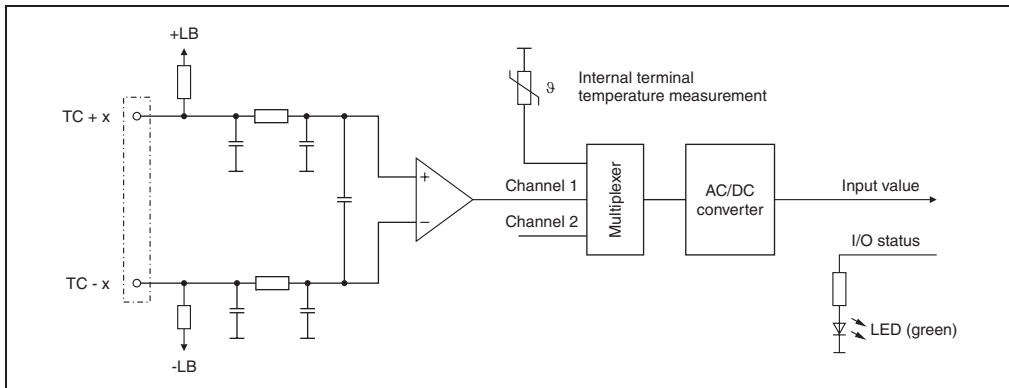


Figure 144: AT2402 - input circuit diagram

12.4.9 Analog inputs

The converted analog values are output by the module in the registers. The sensor type configured will affect the value ranges.

12.4.10 Raw value measurement

If a sensor type other than J, K, N, or S, then the terminal temperature must be measured on at least one input. On the basis of this value, the user must perform a terminal temperature compensation.

12.4.11 Timing setting

The timing setting for data acquisition is made using the converter hardware. All switched on inputs are converted during each conversion cycle. A terminal temperature measurement also takes place (not in function model 1).

If an input is not necessary, it can be switched off, thereby reducing the refresh time. Inputs can also be only temporarily switched off. The measurement of the terminal temperature is switched off in function model 1.

12.4.12 Conversion time

The conversion time depends on the number of channels and on the function model. For the formulas listed in the table, 'n' corresponds to the number of channels that are switched on.

Function model	Conversion time
Model 0 - n channels	$(n + 1) \cdot (2 \cdot \text{FilterTime} + 200\mu\text{s})$
Model 1 - n channels	$n \cdot (2 \cdot \text{FilterTime} + 200\mu\text{s})$
Model 1 - 1 channel	Corresponds to the filter time

Examples

The inputs are filtered using a 50 Hz filter.

	Example 1		Example 2	
	Function model 0	Function model 1	Function model 0	Function model 1
Switched on inputs	1	1	1 + 2	1 + 2
Input conversion times	40.2 ms	20 ms	80.4 ms	80.4 ms
Conversion time for the terminal temperature	40.2 ms	-	40.2 ms	-
Total conversion time	80.4 ms	20 ms	120.6 ms	80.4 ms

12.4.13 Compensation temperature

The module's compensation temperatures can be read.

12.4.14 Setting the external compensation temperature

For each module you can set the compensation temperature used for measurement correction.

12.4.15 Input filter

The filter time for all analog inputs is defined using the input filter parameter.

Value	Filter	Filter time	Digital converter resolution
0	15 Hz	66.7 ms	16-bit
1	25 Hz	40 ms	16-bit
2	30 Hz	33.3 ms	16-bit
3	50 Hz	20 ms	16-bit
4	60 Hz	16.7 ms	16-bit
5	100 Hz	10 ms	16-bit
6	500 Hz	2 ms	16-bit
7	1000 Hz	1 ms	16-bit

12.4.16 Sensor type

The module is designed for various sensor types. The sensor type must be specified because of the different adjustment values.

Code	Input signal
0	Conversion switched off
1	Sensor type J
2	Sensor type K
3	Sensor type S
4	Sensor type N
5	Conversion switched off
6	Raw value without linearization and terminal temperature compensation Resolution 1 μ V for a measurement range of ± 32.767 mV
7	Raw value without linearization and terminal temperature compensation Resolution 2 μ V for a measurement range of ± 65.534 mV

12.4.17 Channel deactivation

The default setting for all channels is ON. To save time, individual channels can be switched off (see section 12.4.12 "Conversion time" on page 345).

Code	Description
0	Channel x: Switched off
1	Channel x: Switched on

12.4.18 Input status

The module's inputs are monitored. A change in the monitoring status generates an error message.

Code	Channel x
0	No error
1	Below lower limit value
2	Above upper limit value
3	Wire break

12.4.19 "StatusInput01" register

Bit	Description
0 - 1	Channel 1: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break
2 - 3	Channel 2: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break
4 - 7	0

In addition to the status info, the error type also sets the analog value to the following values:

Error type	Digital value for error
Wire break	+32767 (\$7FFF)
Above upper limit value	+32767 (\$7FFF)
Below lower limit value	-32767 (\$8001)
Invalid value	-32768 (\$8000)

12.4.20 IOCyclicCounter

The cyclic counter increases after all input data have been updated.

12.4.21 B&R ID code

Code for module identification (\$1BA8).

12.4.22 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
In each operating mode and function model	≥150 µs

12.4.23 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

For the formulas listed in the table, 'n' corresponds to the number of channels that are switched on.

Minimum I/O update time - function model 0	
n inputs	$(n + 1) \cdot (\text{FilterTime} + 200\mu\text{s})$

Minimum I/O update time - function model 1	
1 input	Equal to the filter time
n inputs	$n \cdot (\text{FilterTime} + 200\mu\text{s})$

12.5 AT6402

12.5.1 General information

The AT6402 module is equipped with six inputs for J, K, N and S thermocouple sensors. The module has an integrated terminal temperature compensation.

- 6 inputs for thermocouples
- For sensor types J, K, N, S
- Additional direct raw value measurement
- Integrated terminal temperature compensation
- Filter time can be configured

12.5.2 Order data

Model number	Short description	Image
	Temperature module	
X20AT6402	X20 temperature input module, 6 thermocouple inputs, type J,K,N,S, resolution 0.1 K	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 190: AT6402 - order data

12.5.3 Technical data

Product ID	AT6402
Short description	
I/O module	6 inputs for thermocouples
Thermocouple temperature inputs	
Input	Thermocouple
Digital converter resolution	16-bit
Filter time	Configurable between 1 ms and 66.7 ms
Conversion time 1 channel n channels	80.4 ms at 50 Hz filter (n + 1) x 40.2 ms at 50 Hz filter
Output format	INT
Basic accuracy Type J Type K Type N (Rev. ≥D0) Type S	±0.10 % at 25 °C ¹⁾ ±0.11 % at 25 °C ¹⁾ ±0.11 % at 25 °C ¹⁾ ±0.17 % at 25 °C ¹⁾
Measurement area Sensor temperature FeCuNi: Type J NiCrNi: Type K NiCrSi: Type N (Rev. ≥D0) PtRhPt: Type S Terminal temperature Raw value	-210 °C to +1200 °C -270 °C to +1372 °C -270 °C to +1300 °C -50 °C to +1768 °C 0 °C to +85 °C ±65.534 mV
Terminal temperature compensation	Internal
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Inputs	Yes, with status LED and software status Yes, with status LED and software status
Electrical isolation Channel - Bus Channel - Channel	Yes No
Power consumption Bus I/O internal	Typ. 0.01 W Typ. 0.91 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical

Table 191: AT6402 - technical data

Product ID	AT6402
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 191: AT6402 - technical data (cont.)

1) Based on the measurement range, without consideration of the reference junction measurement error.

12.5.4 Additional technical data

Product ID	AT6402
Thermocouple temperature inputs	
Sensor standard	IEC/653/225/DIS
Resolution Sensor temperature Terminal temperature Raw value output with respect to gain	1 LSB = 0.1 °C 1 LSB = 0.1 °C 1 LSB = 1 µV or 2 µV Gain setting - see section 12.5.16 "Sensor type" on page 356
Standardization Type J Type K Type N (Rev. ≥D0) Type S Terminal temperature	-210.0 °C to +1200.0 °C -270.0 °C to +1372.0 °C -270.0 °C to +1300.0 °C -50.0 °C to +1768.0 °C -25.0 °C to +85.0 °C
Monitoring Open inputs Wire break Below lower range limit Above upper range limit General error	\$7FFF \$7FFF \$8001 \$7FFF \$8000
Conversion method	Sigma Delta
Linearization method	Internal
Input filter	Low pass 1st order / cut-off frequency 500 Hz
Maximum gain drift	<0.0247 % per °C, based on the current measurement value
Maximum offset drift Type J Type K Type N (Rev. ≥D0) Type S	<0.0025 % per °C, based on the entire measurement range <0.0031 % per °C, based on the entire measurement range TBD <0.0092 % per °C, based on the entire measurement range

Table 192: AT6402 - additional technical data

Product ID	AT6402
Terminal temperature compensation precision with natural convection with artificial convection	± 2 °C after 10 min ± 4 °C after 10 min
Common-mode rejection DC 50 Hz	>77 dB >77 dB
Synchronized zone	± 15 V
Cross-talk between channels	-70 dB
Non-linearity	TBD
Isolation voltage betw. channel and bus	500 V _{eff}
General information	
B&R ID code	\$1BA9

Table 192: AT6402 - additional technical data (cont.)

12.5.5 Status LEDs

Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
			Single flash	Warning/error for an I/O channel. Overflow or underflow of the analog inputs.
	e + r	Steady red / single green flash		Invalid firmware
	1 - 6	Green	Off	The input is switched off
			Blinking	Overflow, underflow or broken connection
			On	The analog/digital converter is running

Table 193: AT6402 - status indicators

12.5.6 Pin assignments

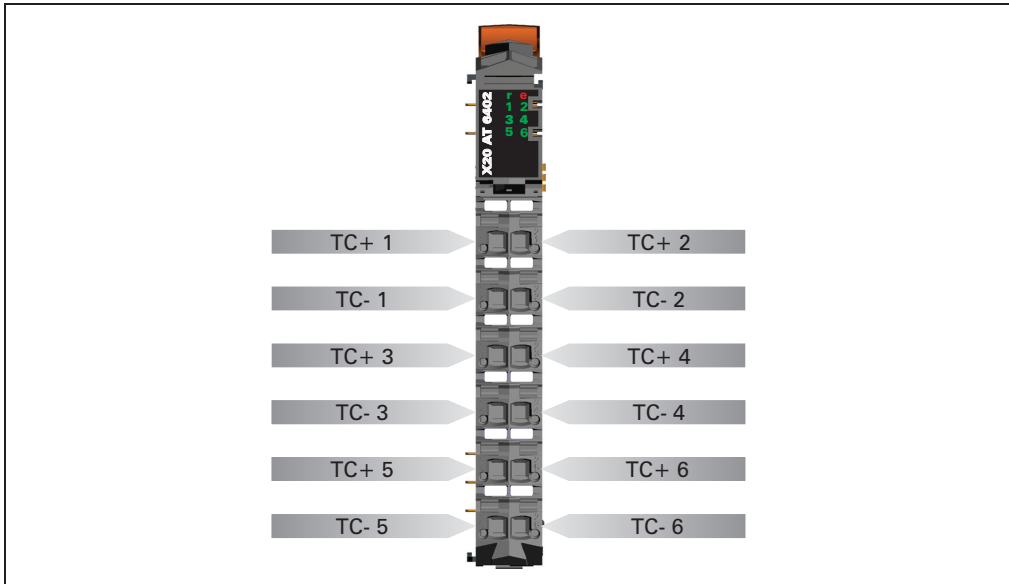


Figure 145: AT6402 - pin assignments

12.5.7 Connection example

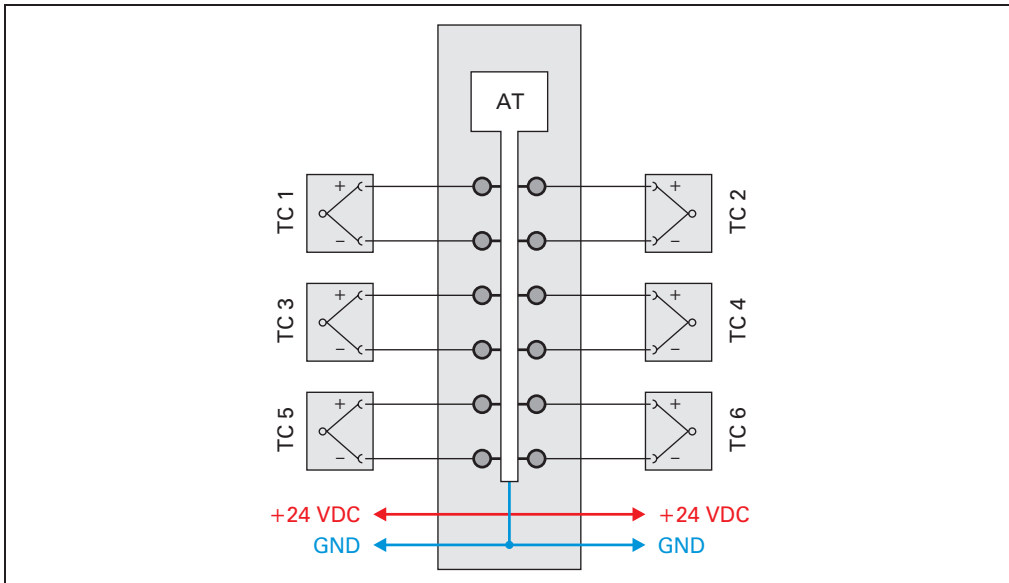


Figure 146: AT6402 - connection example

12.5.8 Input circuit diagram

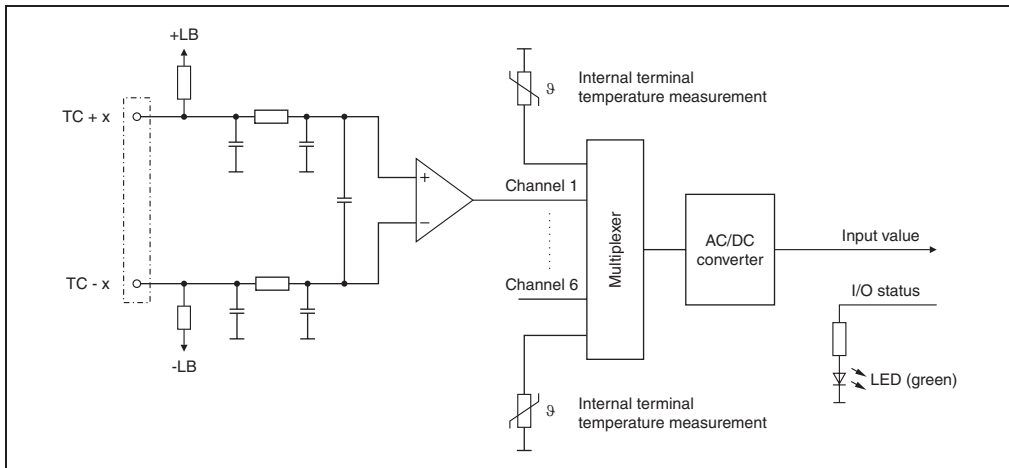


Figure 147: AT6402 - input circuit diagram

12.5.9 Analog inputs

The converted analog values are output by the module in the registers. The sensor type configured will affect the value ranges.

12.5.10 Raw value measurement

If a sensor type other than J, K, N, or S, then the terminal temperature must be measured on at least one input. On the basis of this value, the user must perform a terminal temperature compensation.

12.5.11 Timing setting

The timing setting for data acquisition is made using the converter hardware. All switched on inputs are converted during each conversion cycle. A terminal temperature measurement also takes place (not in function model 1).

If an input is not necessary, it can be switched off, thereby reducing the refresh time. Inputs can also be only temporarily switched off. The measurement of the terminal temperature is switched off in function model 1.

12.5.12 Conversion time

The conversion time depends on the number of channels and on the function model. For the formulas listed in the table, 'n' corresponds to the number of channels that are switched on.

Function model	Conversion time
Model 0 - n channels	$(n + 1) \cdot (2 \cdot \text{FilterTime} + 200\mu\text{s})$
Model 1 - n channels	$n \cdot (2 \cdot \text{FilterTime} + 200\mu\text{s})$
Model 1 - 1 channel	Equal to the filter time

Examples

The inputs are filtered using a 50 Hz filter.

	Example 1		Example 2	
	Function model 0	Function model 1	Function model 0	Function model 1
Switched on inputs	1	1	1 - 6	1 - 6
Input conversion times	40.2 ms	20 ms	241.2 ms	241.2 ms
Conversion time for the terminal temperature	40.2 ms	-	40.2 ms	-
Total conversion time	80.4 ms	20 ms	281.4 ms	241.2 ms

12.5.13 Compensation temperature

The module's compensation temperatures can be read.

12.5.14 Setting the external compensation temperature

For each module you can set the compensation temperature used for measurement correction.

12.5.15 Input filter

The filter time for all analog inputs is defined using the input filter parameter.

Value	Filter	Filter time	Digital converter resolution
0	15 Hz	66.7 ms	16-bit
1	25 Hz	40 ms	16-bit
2	30 Hz	33.3 ms	16-bit
3	50 Hz	20 ms	16-bit
4	60 Hz	16.7 ms	16-bit
5	100 Hz	10 ms	16-bit
6	500 Hz	2 ms	16-bit
7	1000 Hz	1 ms	16-bit

12.5.16 Sensor type

The module is designed for various sensor types. The sensor type must be specified because of the different adjustment values.

Code	Input signal
0	Conversion switched off
1	Sensor type J
2	Sensor type K
3	Sensor type S
4	Sensor type N
5	Conversion switched off
6	Raw value without linearization and terminal temperature compensation Resolution 1 μ V for a measurement range of ± 32.767 mV
7	Raw value without linearization and terminal temperature compensation Resolution 2 μ V for a measurement range of ± 65.534 mV

12.5.17 Channel deactivation

The default setting for all channels is ON. To save time, individual channels can be switched off (see section 12.5.12 "Conversion time" on page 355).

Code	Description
0	Channel x: Switched off
1	Channel x: Switched on

12.5.18 Input status

The module's inputs are monitored. A change in the monitoring status generates an error message.

Code	Channel x
0	No error
1	Below lower limit value
2	Above upper limit value
3	Wire break

12.5.19 "StatusInput01" register

Bit	Description
0 - 1	Channel 1: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break
2 - 3	Channel 2: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break
4 - 5	Channel 3: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break
6 - 7	Channel 4: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break

12.5.20 "StatusInput02" register

Bit	Description
0 - 1	Channel 5: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break
2 - 3	Channel 6: 00 ... No error 01 ... Below lower limit value 10 ... Above upper limit value 11 ... Wire break
4 - 7	0

In addition to the status info, the error type also sets the analog value to the following values:

Error type	Digital value for error
Wire break	+32767 (\$7FFF)
Above upper limit value	+32767 (\$7FFF)
Below lower limit value	-32767 (\$8001)
Invalid value	-32768 (\$8000)

12.5.21 IOcyclicCounter

The cyclic counter increases after all input data have been updated.

12.5.22 B&R ID code

Code for module identification (\$1BA9).

12.5.23 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
In each operating mode and function model	$\geq 150 \mu\text{s}$

12.5.24 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

For the formulas listed in the table, 'n' corresponds to the number of channels that are switched on.

Minimum I/O update time - function model 0	
n inputs	$(n + 1) \cdot (\text{FilterTime} + 200\mu\text{s})$

Minimum I/O update time - function model 1	
1 input	Equal to the filter time
n inputs	$n \cdot (\text{FilterTime} + 200\mu\text{s})$

13. Counter modules

13.1 General information

Counter modules are used for position detection. Each signal on a counter module is assigned to a status LED.

13.1.1 Overview

Model number	Short description
X20DC1196	X20 digital counter module, 1 channel ABR, 5 V, 250 kHz input freq., 4x evaluation
X20DC1198	X20 digital counter module, 1 channel SSI, 5 V, 1 MBit/s, 32-bit
X20DC1396	X20 digital counter module, 1 channel ABR, 24 V, 100 kHz input freq., 4x evaluation
X20DC1398	X20 digital counter module, 1 channel SSI, 24 V, 125 kBit/s, 32-bit
X20DC2396	X20 digital counter module, 2 ABR channels, 24 V, 100 kHz input freq., 4x evaluation
X20DC2398	X20 digital counter module, 2 SSI channels, 24 V, 125 kBit/s, 32-bit
X20DC2395	X20 digital counter module, 1 x SSI absolute encoder (24 V), 1 x ABR incremental encoder (24 V), 2 x AB incremental encoder (24 V), 4 x event counter, or 2 x PWM
X20DC4395	X20 digital counter module, 2 x SSI absolute encoder (24 V), 2 x ABR incremental encoder (24 V), 4 x AB incremental encoder (24 V), 8 x event counter, or 4 x PWM

Table 194: Overview of counter modules

13.2 DC1196

13.2.1 General information

The DC1196 module is equipped with an input for ABR incremental encoder with 5 V encoder signal.

- One 5 V ABR incremental encoder
- 2 additional inputs e.g. for reference enable switch
- 5 VDC, 24 VDC and GND for encoder supply

13.2.2 Order data

Model number	Short description	Image
	Counter module	
X20DC1196	X20 digital counter module, 1 channel ABR, 5 V, 250 kHz input freq., 4x evaluation	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 195: DC1196 - order data

13.2.3 Technical data

Product ID	DC1196
Short description	
I/O module	One 5 V ABR incremental encoder
ABR incremental encoder	
Encoder inputs	5 V, symmetrical
Counter size	16-bit
Input frequency (max.)	250 kHz

Table 196: DC1196 - technical data

Product ID	DC1196
Evaluation	4-fold
Encoder supply 5 V 24 V	Module-internal, max. 300 mA Module-internal, max. 300 mA
Digital inputs	
Number	2
Rated voltage	24 VDC
Input filter Hardware Software	≤2 µs -
Input circuit	Sink
Additional functions	Reference enable switch
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error	Yes, with status LED and software status
Electrical isolation Encoder - Bus Channel - Bus Channel - Encoder Channel - Channel	Yes Yes No No
Power consumption Bus I/O internal	Typ. 0.01 W Typ. 1.5 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 196: DC1196 - technical data (cont.)

13.2.4 Additional technical data

Product ID	DC1196
ABR incremental encoder	
Input filter Hardware Software	$\leq 200 \text{ ns}$ -
Synchronized zone	$-7 \text{ V} \leq V_{\text{CM}} \leq +12 \text{ V}$
Overload behavior of the encoder supply	Short circuit protection, overload protection
Isolation voltage betw. encoder and bus	$500 \text{ V}_{\text{eff}}$
Digital inputs	
Input voltage	$24 \text{ VDC} (-15 \% / +20 \%)$
Input current at 24 VDC	Approx. 3.3 mA
Input resistance	$7.19 \text{ k}\Omega$
Switching threshold Low High	$< 5 \text{ VDC}$ $> 15 \text{ VDC}$
Isolation voltage betw. channel and bus	$500 \text{ V}_{\text{eff}}$
General information	
B&R ID code	\$1BAF
Type of signal lines	Shielded cables should be used for all signal lines

Table 197: DC1196 - additional technical data

13.2.5 Status LEDs

Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
	A1	Green		Input state of counter input A
	B1	Green		Input state of counter input B
	R1	Green		Input state of reference pulse R
	1 - 2	Green		Input status of the corresponding digital input

Table 198: DC1196 - status indicators

13.2.6 Pin assignments

Shielded cables should be used for all signal lines.

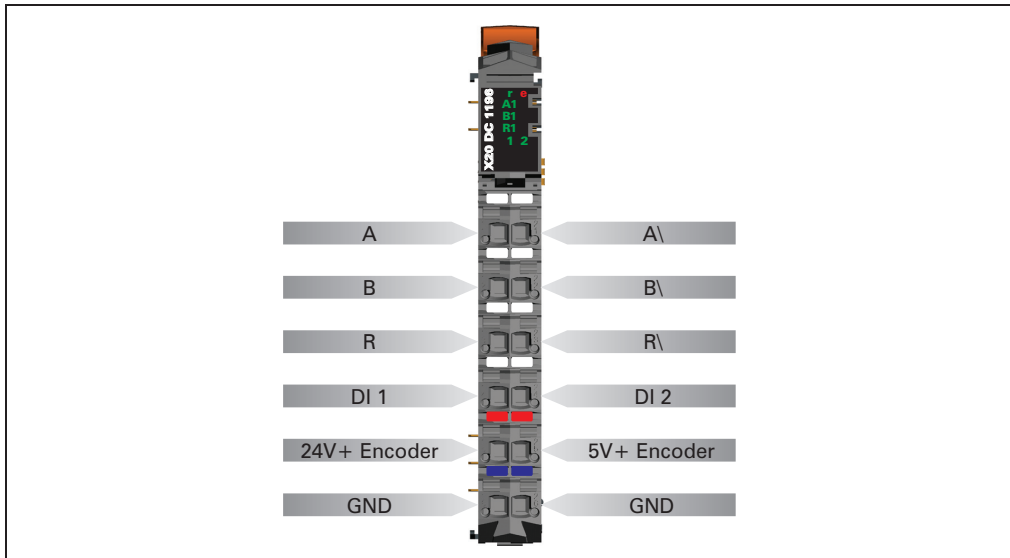


Figure 148: DC1196 - pin assignments

13.2.7 Connection example

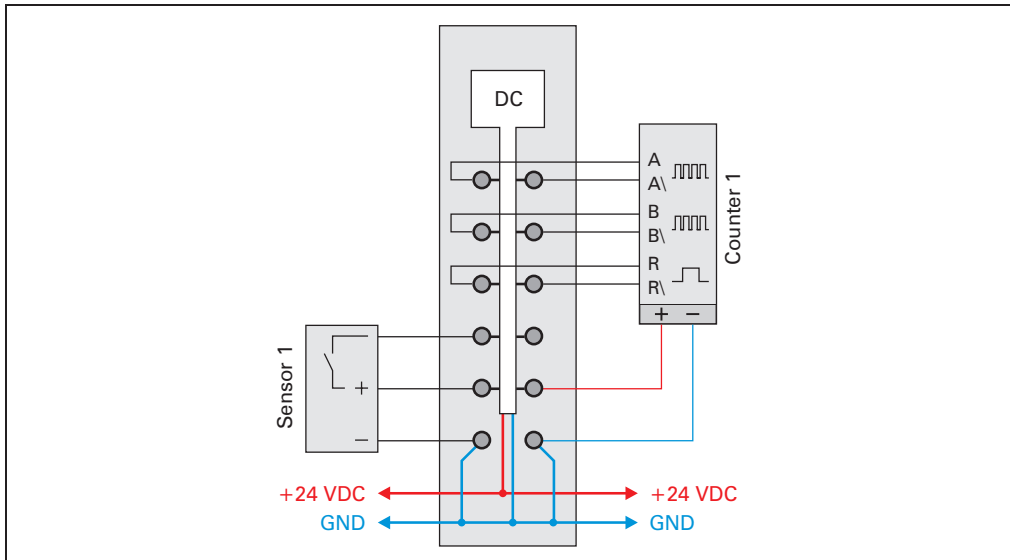


Figure 149: DC1196 - connection example

13.2.8 Input circuit diagram

Counter inputs

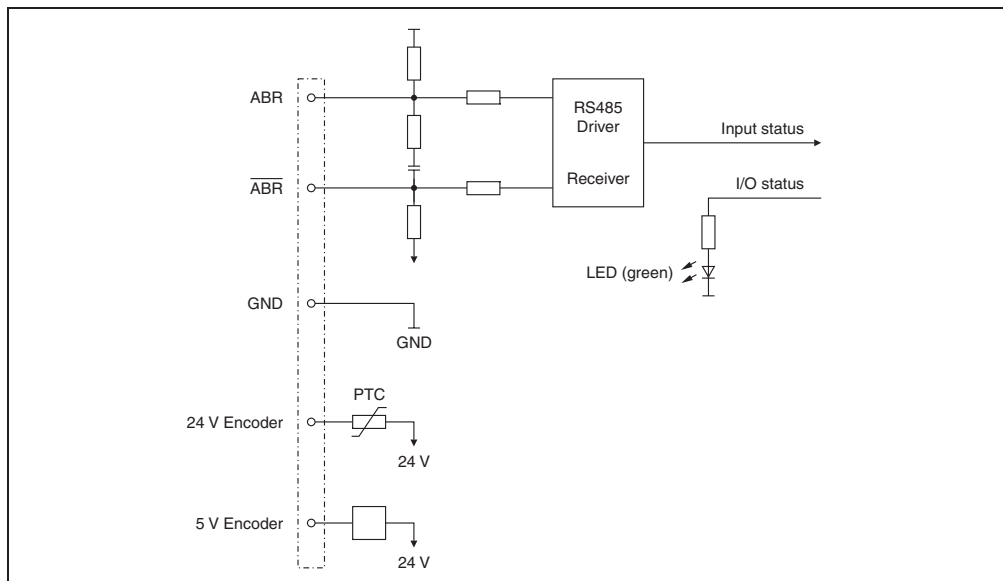


Figure 150: DC1196 - input circuit diagram - counter inputs

Standard inputs

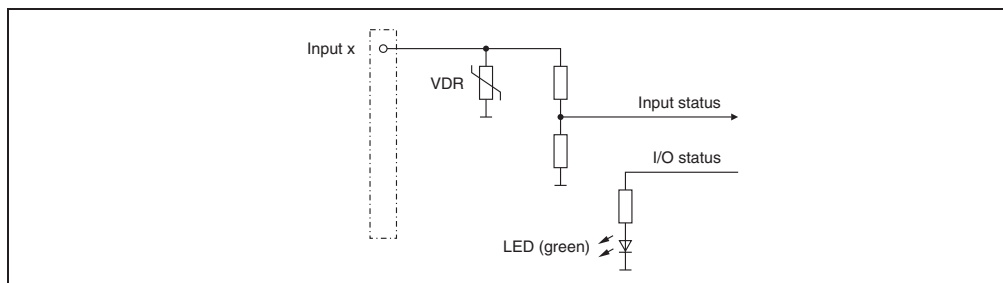


Figure 151: DC1196 - input circuit diagram - standard inputs

13.2.9 ReferenceModeEncoder01 channel

The reference mode is written cyclically by incremental encoder 1. Bit 0 and bit 1 make up the definition:

- \$C1 ... Single reference
- \$C3 ... Continual references

After a successful single reference (indicated by the status bits), the reference mode must first be turned back off (value = \$C0).

After the status has taken on the value \$00, the reference process can be started again with the value \$C1.

Bit	Description
0 - 1	Setting the reference mode 00 ... Off (default) 01 ... Single reference 10 ... Reserved 11 ... Continual references
2 - 5	0
6 - 7	1

13.2.10 StatusInput01 channel

The status of incremental encoder 1 is read cyclically.

Bit	Description
0 - 1	0
2	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
3	Toggles after every completed reference 0 ... Referencing is turned off 0/1 ... Toggle after each completed reference
4	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
5 - 7	Cyclic counter which is increased by 1 as soon as a reference pulse is detected 0 ... Referencing is turned off xxx ... Counter status

13.3 DC1198

13.3.1 General information

The DC1198 module is equipped with an input for SSI absolute encoder with 5 V encoder signal.

- 1 5 V SSI absolute encoder
- 2 additional inputs
- 5 VDC, 24 VDC and GND for encoder supply

13.3.2 Order data

Model number	Short description	Image
	Counter module	
X20DC1198	X20 digital counter module, 1 channel SSI, 5 V, 1 MBit/s, 32-bit	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 199: DC1198 - order data

13.3.3 Technical data

Product ID	DC1198
Short description	
I/O module	1 5 V SSI absolute encoder
SSI absolute encoder	
Encoder inputs	5 V, symmetrical
Counter size	32-bit
Maximum transfer rate	1 MBit/s

Table 200: DC1198 - technical data

Product ID	DC1198
Encoder supply 5 V 24 V	Module-internal, max. 300 mA Module-internal, max. 300 mA
Digital inputs	
Number	2
Rated voltage	24 VDC
Input filter Hardware Software	≤2 µs -
Input circuit	Sink
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error	Yes, with status LED and software status
Electrical isolation Encoder - Bus Channel - Bus Channel - Encoder Channel - Channel	Yes Yes No No
Power consumption Bus I/O internal	Typ. 0.01 W Typ. 1.5 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 200: DC1198 - technical data (cont.)

13.3.4 Additional technical data

Product ID	DC1198
SSI absolute encoder	
Coding	Gray/Binary
Transfer rate	125 kBit/s / 250 kBit/s / 500 kBit/s / 1 MBit/s
Overload behavior of the encoder supply	Short-circuit protection, overload protection
Isolation voltage betw. encoder and bus	500 V _{eff}
Digital inputs	
Input voltage	24 VDC (-15 % / +20 %)
Input current at 24 VDC	Approx. 3.3 mA
Input resistance	7.19 kΩ
Switching threshold Low High	<5 VDC >15 VDC
Isolation voltage betw. channel and bus	500 V _{eff}
General information	
B&R ID code	\$1BB0
Type of signal lines	Shielded cables should be used for all signal lines

Table 201: DC1198 - additional technical data

13.3.5 Status LEDs

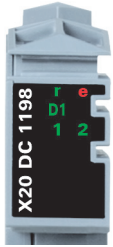
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
	D1	Green		Input state of data signal
	1 - 2	Green		Input status of the corresponding digital input

Table 202: DC1198 - status indicator

13.3.6 Pin assignments

Shielded cables should be used for all signal lines.

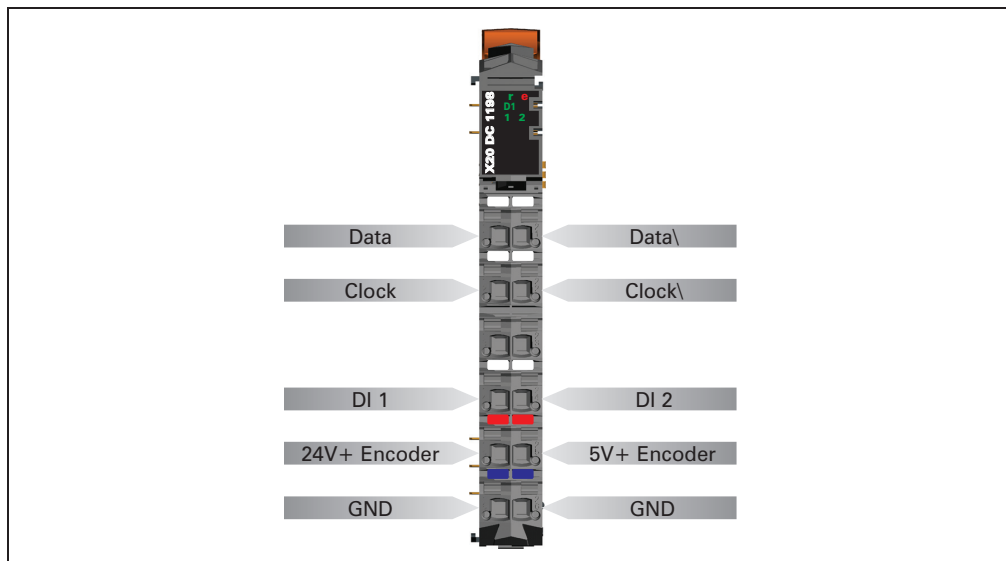


Figure 152: DC1198 - pin assignments

13.3.7 Connection example

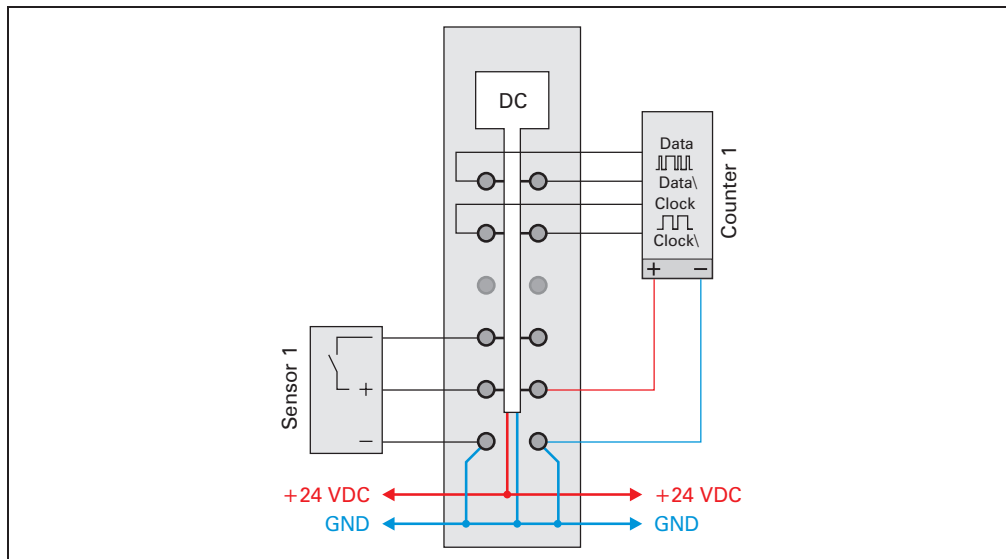


Figure 153: DC1198 - connection example

13.3.8 Input circuit diagram

Counter input

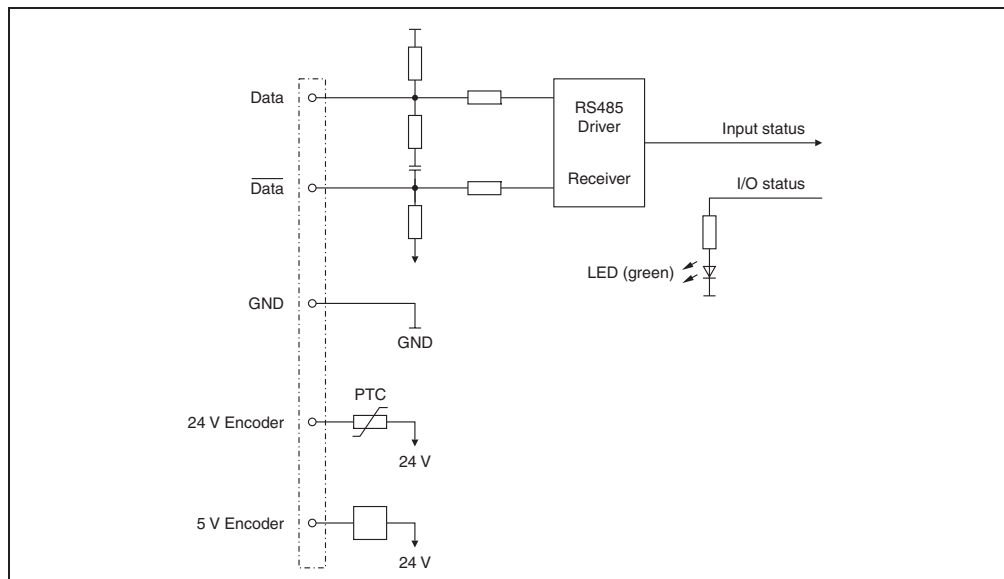


Figure 154: DC1198 - input circuit diagram - counter inputs

Standard inputs

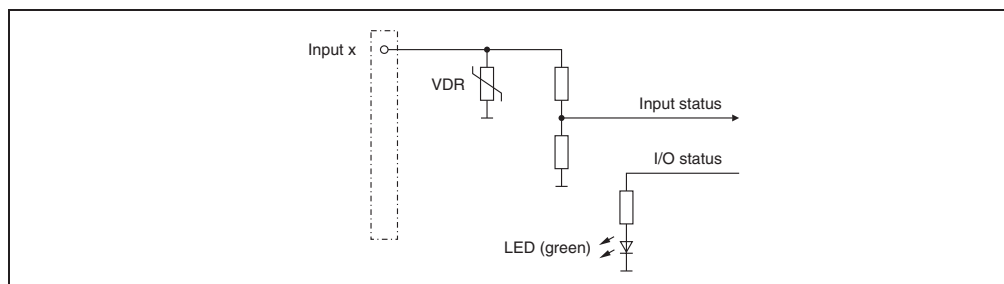


Figure 155: DC1198 - input circuit diagram - standard inputs

13.3.9 Output circuit diagram

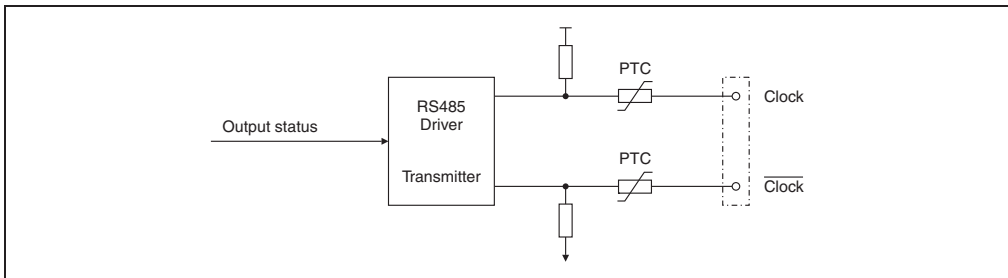


Figure 156: DC1198 - output circuit diagram

13.4 DC1396

13.4.1 General information

The DC1396 module is equipped with an input for ABR incremental encoder with 24 V encoder signal.

- One 24 V ABR incremental encoder
- 1 additional input e.g. for reference enable switch
- 24 VDC and GND for encoder supply

13.4.2 Order data

Model number	Short description	Image
	Counter module	
X20DC1396	X20 digital counter module, 1 channel ABR, 24 V, 100 kHz input freq., 4x evaluation	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 203: DC1396 - order data

13.4.3 Technical data

Product ID	DC1396
Short description	
I/O module	One 24 V ABR incremental encoder
ABR incremental encoder	
Encoder inputs	24 V, asymmetrical
Counter size	16-bit
Input frequency (max.)	100 kHz

Table 204: DC1396 - technical data

Product ID	DC1396
Evaluation	4-fold
Encoder supply	Module-internal, max. 600 mA
Reference enable switch	
Number	1
Rated voltage	24 VDC
Input filter Hardware Software	$\leq 2 \mu\text{s}$ -
Input circuit	Sink
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error	Yes, with status LED and software status
Electrical isolation Encoder - Bus Reference enable switch - Bus Reference enable switch - Encoder	Yes Yes No
Power consumption Bus I/O internal	Typ. 0.01 W Typ. 1.4 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 204: DC1396 - technical data (cont.)

13.4.4 Additional technical data

Product ID	DC1396
ABR incremental encoder	
Input filter Hardware Software	$\leq 2 \mu s$ -
Input current at 24 VDC	Approx. 1.3 mA
Input resistance	18.4 k Ω
Switching threshold Low High	<5 VDC >15 VDC
Overload behavior of the encoder supply	Short circuit protection, overload protection
Isolation voltage betw. encoder and bus	500 V _{eff}
Reference enable switch	
Input current at 24 VDC	Approx. 3.3 mA
Input resistance	7.19 k Ω
Switching threshold Low High	<5 VDC >15 VDC
Isolation voltage between reference enable switch and bus	500 V _{eff}
General information	
B&R ID code	\$1BAC
Input voltage	24 VDC (-15 % / +20 %)
Type of signal lines	Shielded cables should be used for all signal lines

Table 205: DC1396 - additional technical data

13.4.5 Status LEDs

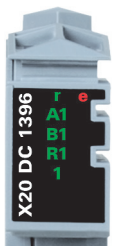
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
	A1	Green		Input state of counter input A
	B1	Green		Input state of counter input B
	R1	Green		Input state of reference pulse R
	1	Green		Input state - digital input

Table 206: DC1396 - status indicators

13.4.6 Pin assignments

Shielded cables should be used for all signal lines.

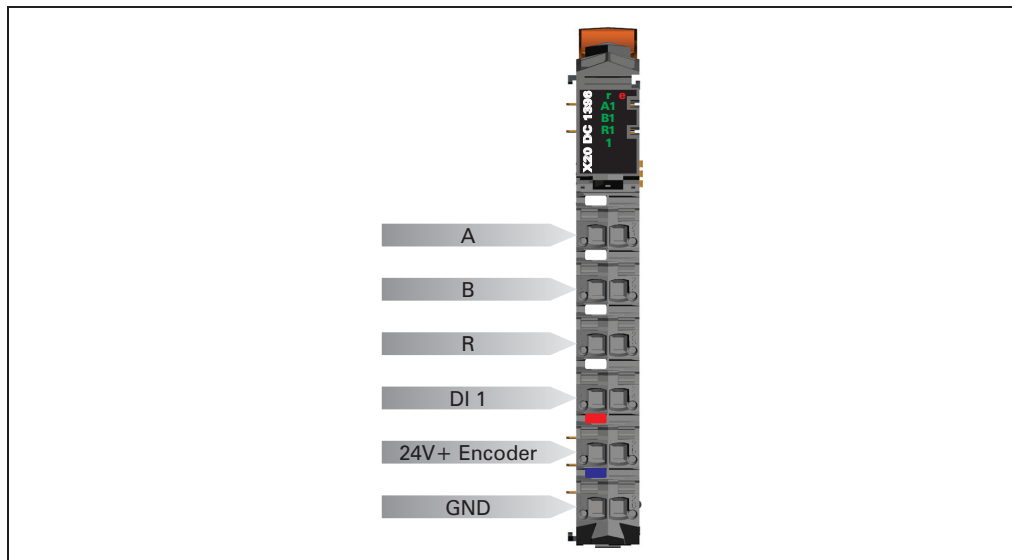


Figure 157: DC1396 - pin assignments

13.4.7 Connection example

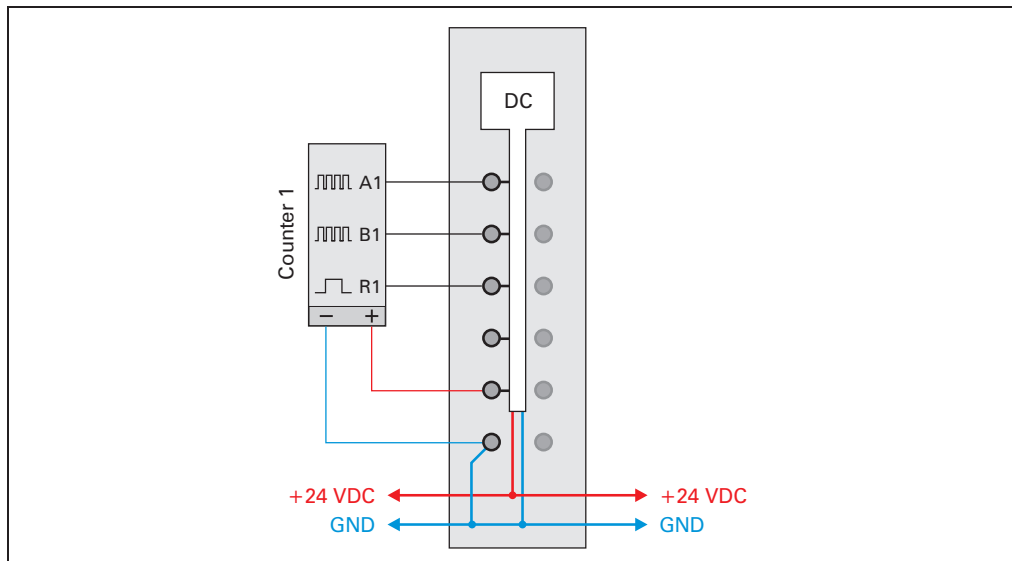


Figure 158: DC1396 - connection example

13.4.8 Input circuit diagram

Counter inputs

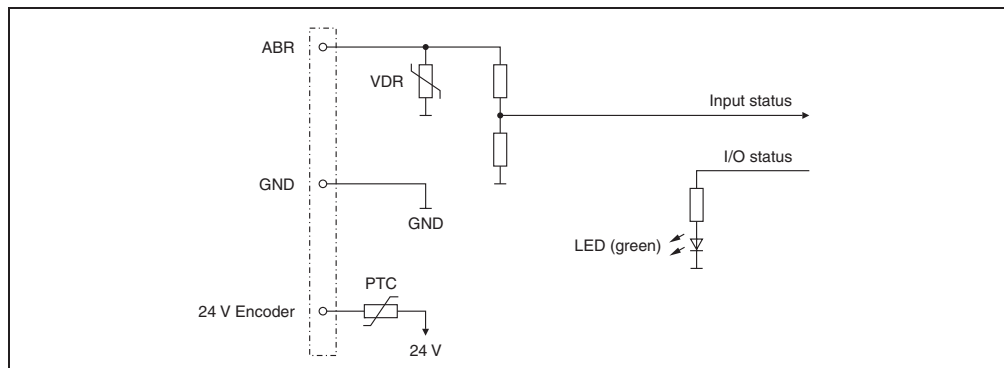


Figure 159: DC1396 - input circuit diagram - counter inputs

Standard input

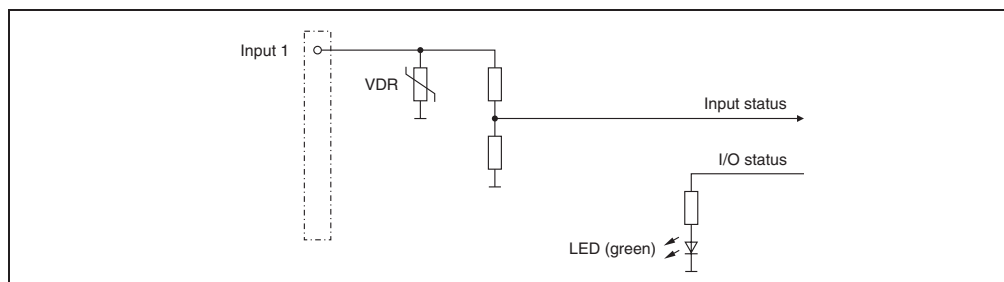


Figure 160: DC1396 - input circuit diagram - standard input

13.4.9 ReferenceModeEncoder01 channel

The reference mode is written cyclically by incremental encoder 1. Bit 0 and bit 1 make up the definition:

- \$C1 ... Single reference
- \$C3 ... Continual references

After a successful single reference (indicated by the status bits), the reference mode must first be turned back off (value = \$C0).

After the status has taken on the value \$00, the reference process can be started again with the value \$C1.

Bit	Description
0 - 1	Setting the reference mode 00 ... Off (default) 01 ... Single reference 10 ... Reserved 11 ... Continual references
2 - 5	0
6 - 7	1

13.4.10 StatusInput01 channel

The status of incremental encoder 1 is read cyclically.

Bit	Description
0 - 1	0
2	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
3	Toggles after every completed reference 0 ... Referencing is turned off 0/1 ... Toggle after each completed reference
4	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
5 - 7	Cyclic counter which is increased by 1 as soon as a reference pulse is detected 0 ... Referencing is turned off xxx ... Counter status

13.5 DC1398

13.5.1 General information

The DC1398 module is equipped with an input for SSI absolute encoder with 24 V encoder signal.

- 1 24 V SSI absolute encoder
- 1 additional input
- 24 VDC and GND for encoder supply

13.5.2 Order data

Model number	Short description	Image
	Counter module	
X20DC1398	X20 digital counter module, 1 channel SSI, 24 V, 125 kBit/s, 32-bit	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 207: DC1398 - order data

13.5.3 Technical data

Product ID	DC1398
Short description	
I/O module	1 24 V SSI absolute encoder
SSI absolute encoder	
Encoder inputs	24 V, asymmetrical
Counter size	32-bit

Table 208: DC1398 - technical data

Product ID	DC1398
Maximum transfer rate	125 kBit/s
Encoder supply	Module-internal, max. 600 mA
Digital inputs	
Number	1
Rated voltage	24 VDC
Input filter Hardware Software	≤2 µs -
Input circuit	Sink
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error	Yes, with status LED and software status
Electrical isolation Encoder - Bus Channel - Bus Channel - Encoder	Yes Yes No
Power consumption Bus I/O internal	Typ. 0.01 W Typ. 1.3 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 208: DC1398 - technical data (cont.)

13.5.4 Additional technical data

Product ID	DC1398
SSI absolute encoder	
Coding	Gray/Binary
DATA: Input resistance	18.4 k Ω
CLK: Output current	Max. 100 mA
Switching threshold Low High	<5 VDC >15 VDC
Overload behavior of the encoder supply	Short circuit protection, overload protection
Isolation voltage betw. encoder and bus	500 V _{eff}
Digital inputs	
Input current at 24 VDC	Approx. 3.3 mA
Input resistance	7.19 k Ω
Switching threshold Low High	<5 VDC >15 VDC
Isolation voltage betw. channel and bus	500 V _{eff}
General information	
B&R ID code	\$1BAE
Input voltage	24 VDC (-15 % / +20 %)
Type of signal lines	Shielded cables should be used for all signal lines

Table 209: DC1398 - additional technical data

13.5.5 Status LEDs

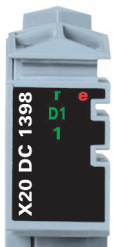
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
	D1	Green		Input state of data signal
	1	Green		Input state - digital input

Table 210: DC1398 - status indicators

13.5.6 Pin assignments

Shielded cables should be used for all signal lines.

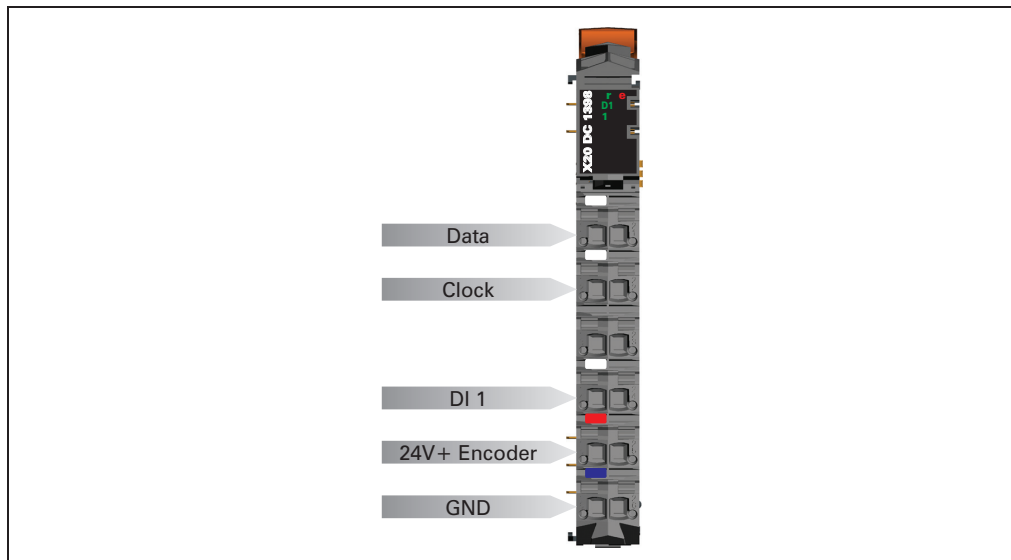


Figure 161: DC1398 - pin assignments

13.5.7 Connection example

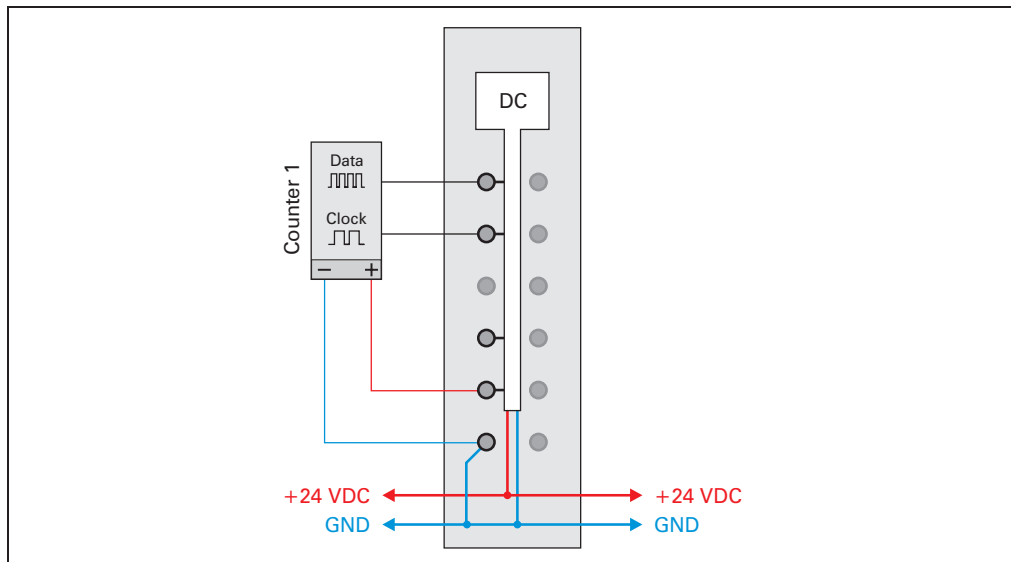


Figure 162: DC1398 - connection example

13.5.8 Input circuit diagram

Counter input

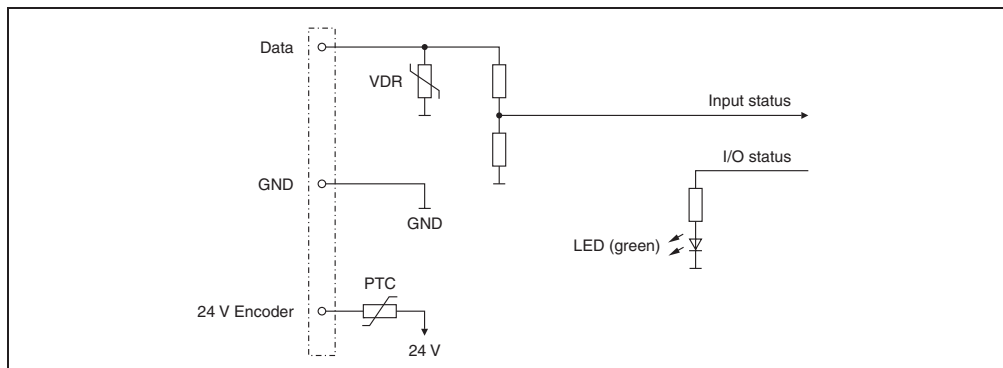


Figure 163: DC1398 - input circuit diagram - counter inputs

Standard input

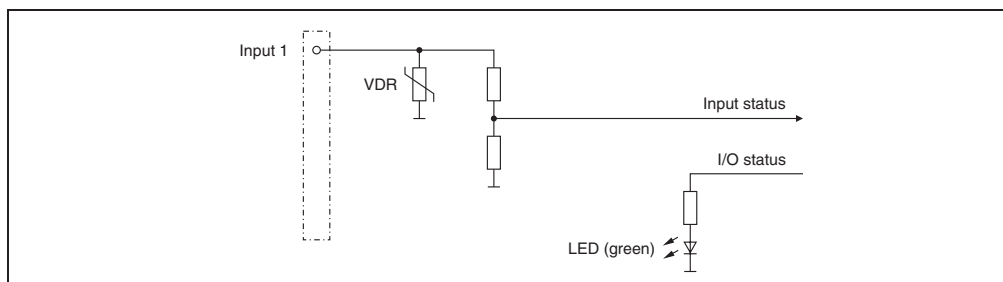


Figure 164: DC1398 - input circuit diagram - standard input

13.5.9 Output circuit diagram

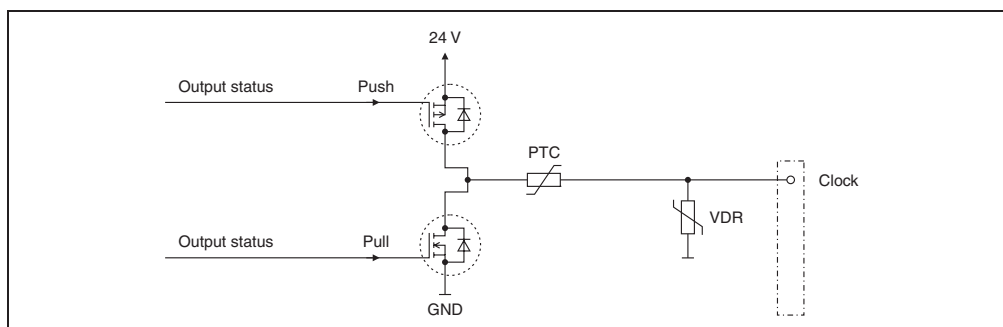


Figure 165: DC1398 - output circuit diagram

13.6 DC2396

13.6.1 General information

The DC2396 module is equipped with two inputs for ABR incremental encoder with 24 V encoder signal.

- Two 24 V ABR incremental encoder
- 2 additional inputs e. g. for reference enable switch
- 24 VDC and GND for encoder supply

13.6.2 Order data

Model number	Short description	Image
	Counter module	
X20DC2396	X20 digital counter module, 2 ABR channels, 24 V, 100 kHz input freq., 4x evaluation	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 211: DC2396 - order data

13.6.3 Technical data

Product ID	DC2396
Short description	
I/O module	Two 24 V ABR incremental encoder
ABR incremental encoder	
Encoder inputs	24 V, asymmetrical
Counter size	16-bit
Input frequency (max.)	100 kHz

Table 212: DC2396 - technical data

X20 System modules • Counter modules • DC2396

Product ID	DC2396
Evaluation	4-fold
Encoder supply	Module-internal, max. 600 mA
Reference enable switch	
Number	2
Rated voltage	24 VDC
Input filter Hardware Software	≤2 µs -
Input circuit	Sink
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error	Yes, with status LED and software status
Electrical isolation Encoder - Bus Reference enable switch - Bus Reference enable switch - Encoder Encoder - Encoder Reference switch - Reference switch	Yes Yes No No No
Power consumption Bus I/O internal	Typ. 0.01 W Typ. 1.5 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 212: DC2396 - technical data (cont.)

13.6.4 Additional technical data

Product ID	DC2396
ABR incremental encoder	
Input filter Hardware Software	$\leq 2 \mu\text{s}$ -
Input current at 24 VDC	Approx. 1.3 mA
Input resistance	18.4 k Ω
Switching threshold Low High	<5 VDC >15 VDC
Overload behavior of the encoder supply	Short circuit protection, overload protection
Isolation voltage betw. encoder and bus	500 V _{eff}
Reference enable switch	
Input current at 24 VDC	Approx. 3.3 mA
Input resistance	7.19 k Ω
Switching threshold Low High	<5 VDC >15 VDC
Isolation voltage between reference enable switch and bus	500 V _{eff}
General information	
B&R ID code	\$1BAB
Input voltage	24 VDC (-15 % / +20 %)
Type of signal lines	Shielded cables should be used for all signal lines

Table 213: DC2396 - additional technical data

13.6.5 Status LEDs

Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
	A1, A2	Green		Input state of counter input A1 or A2
	B1, B2	Green		Input state of counter input B1 or B2
	R1, R2	Green		Input state of reference pulse R1 or R2
	1 - 2	Green		Input status of the corresponding digital input

Table 214: DC2396 - status indicators

13.6.6 Pin assignments

Shielded cables should be used for all signal lines.

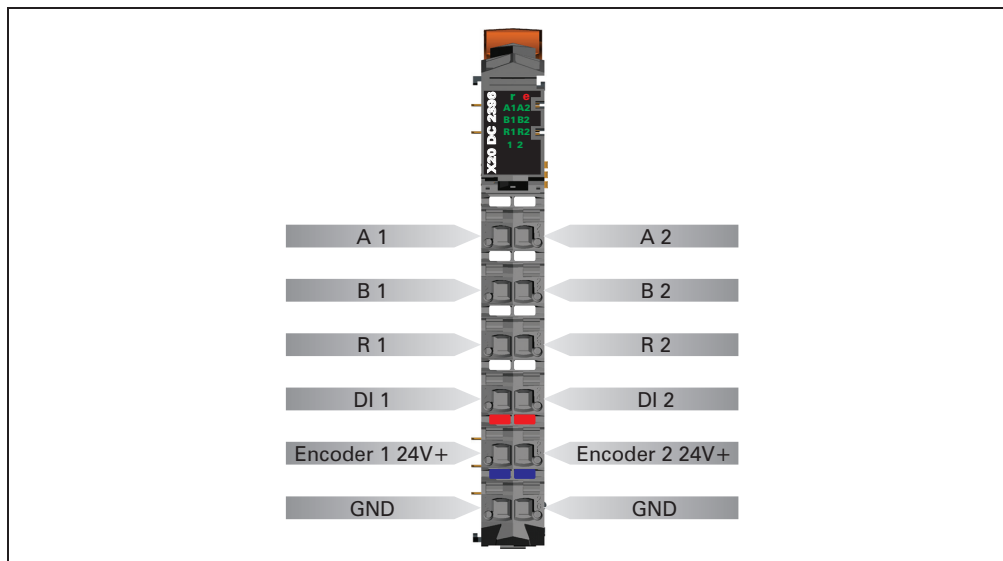


Figure 166: DC2396 - pin assignments

13.6.7 Connection example

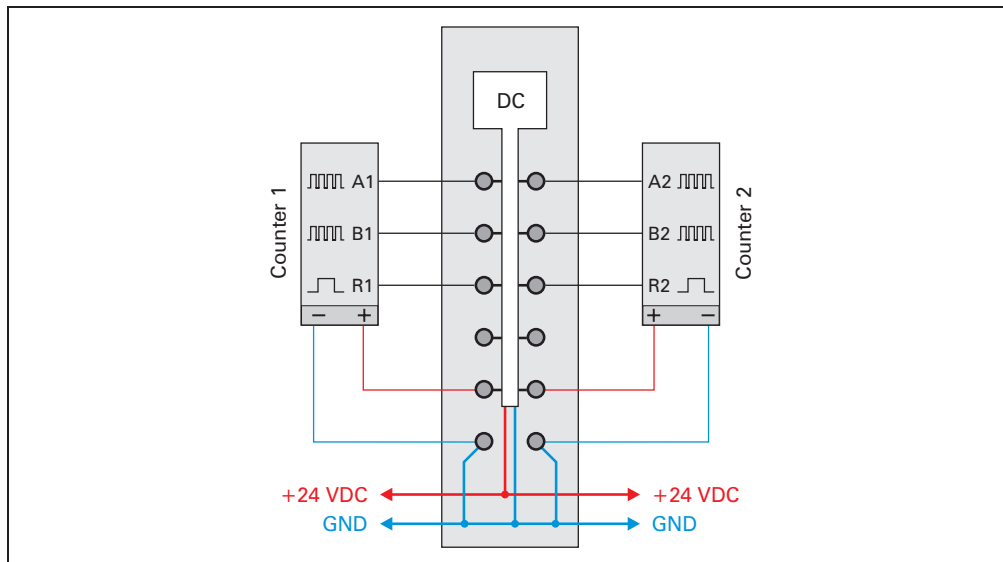


Figure 167: DC2396 - connection example

13.6.8 Input circuit diagram

Counter inputs

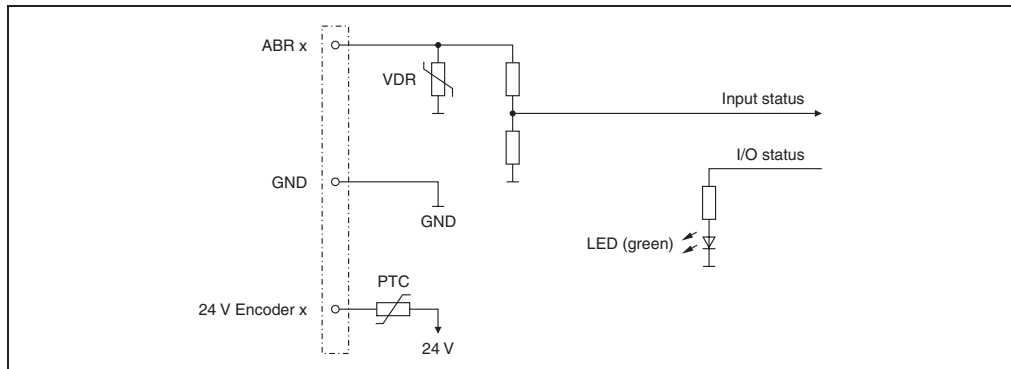


Figure 168: DC2396 - input circuit diagram - counter inputs

Standard inputs

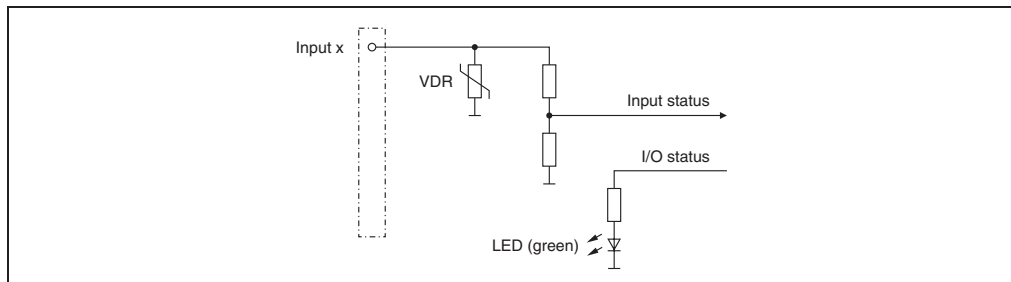


Figure 169: DC2396 - input circuit diagram - standard input

13.6.9 ReferenceModeEncoder01 channel

The reference mode is written cyclically by incremental encoder 1. Bit 0 and bit 1 make up the definition:

\$C1 ... Single reference
\$C3 ... Continual references

After a successful single reference (indicated by the status bits), the reference mode must first be turned back off (value = \$C0).

After the status has taken on the value \$00, the reference process can be started again with the value \$C1.

Bit	Description
0 - 1	Setting the reference mode 00 ... Off (default) 01 ... Single reference 10 ... Reserved 11 ... Continual references
2 - 5	0
6 - 7	1

13.6.10 ReferenceModeEncoder02 channel

The reference mode is written cyclically by incremental encoder 2. Bit 0 and bit 1 make up the definition:

\$C1 ... Single reference
\$C3 ... Continual references

After a successful single reference (indicated by the status bits), the reference mode must first be turned back off (value = \$C0).

After the status has taken on the value \$00, the reference process can be started again with the value \$C1.

Bit	Description
0 - 1	Setting the reference mode 00 ... Off (default) 01 ... Single reference 10 ... Reserved 11 ... Continual references
2 - 5	0
6 - 7	1

13.6.11 StatusInput01 channel

The status of incremental encoder 1 is read cyclically.

Bit	Description
0 - 1	0
2	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
3	Toggles after every completed reference 0 ... Referencing is turned off 0/1 ... Toggle after each completed reference
4	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
5 - 7	Cyclic counter which is increased by 1 as soon as a reference pulse is detected 0 ... Referencing is turned off xxx ... Counter status

13.6.12 StatusInput02 channel

The status of incremental encoder 2 is read cyclically.

Bit	Description
0 - 1	0
2	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
3	Toggles after every completed reference 0 ... Referencing is turned off 0/1 ... Toggle after each completed reference
4	After the first reference pulse, this is always 1 0 ... Referencing is turned off 1 ... After the first reference pulse
5 - 7	Cyclic counter which is increased by 1 as soon as a reference pulse is detected 0 ... Referencing is turned off xxx ... Counter status

13.7 DC2398

13.7.1 General information

The DC2398 module is equipped with two inputs for SSI absolute encoders with a 24 V encoder signal.

- 2 24 V SSI absolute encoder
- 2 additional inputs
- 24 VDC and GND for encoder supply

13.7.2 Order data

Model number	Short description	Image
	Counter module	
X20DC2398	X20 digital counter module, 2 SSI channels, 24 V, 125 kBit/s, 32-bit	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 215: DC2398 - order data

13.7.3 Technical data

Product ID	DC2398
Short description	
I/O module	2 24 V SSI absolute encoder
SSI absolute encoder	
Encoder inputs	24 V, asymmetrical
Counter size	32-bit
Maximum transfer rate	125 kBit/s
Encoder supply	Module-internal, max. 600 mA

Table 216: DC2398 - technical data

Product ID	DC2398
Digital inputs	
Number	2
Rated voltage	24 VDC
Input filter Hardware Software	≤2 µs -
Input circuit	Sink
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error	Yes, with status LED and software status
Electrical isolation Encoder - Bus Channel - Bus Channel - Encoder Encoder - Encoder Channel - Channel	Yes Yes No No No
Power consumption Bus I/O internal	Typ. 0.01 W Typ. 1.4 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 216: DC2398 - technical data (cont.)

13.7.4 Additional technical data

Product ID	DC2398
SSI absolute encoder	
Coding	Gray/Binary
DATA: Input resistance	18.4 k Ω
CLK: Output current	Max. 100 mA
Switching threshold Low High	<5 VDC >15 VDC
Overload behavior of the encoder supply	Short circuit protection, overload protection
Isolation voltage betw. encoder and bus	500 V _{eff}
Digital inputs	
Input current at 24 VDC	Approx. 3.3 mA
Input resistance	7.19 k Ω
Switching threshold Low High	<5 VDC >15 VDC
Isolation voltage betw. channel and bus	500 V _{eff}
General information	
B&R ID code	\$1BAD
Input voltage	24 VDC (-15 % / +20 %)
Type of signal lines	Shielded cables should be used for all signal lines

Table 217: DC2398 - additional technical data

13.7.5 Status LEDs

Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
	D1, D2	Green		Input state of data signal 1 or 2
	1 - 2	Green		Input status of the corresponding digital input

Table 218: DC2398 - status indicators

13.7.6 Pin assignments

Shielded cables should be used for all signal lines.

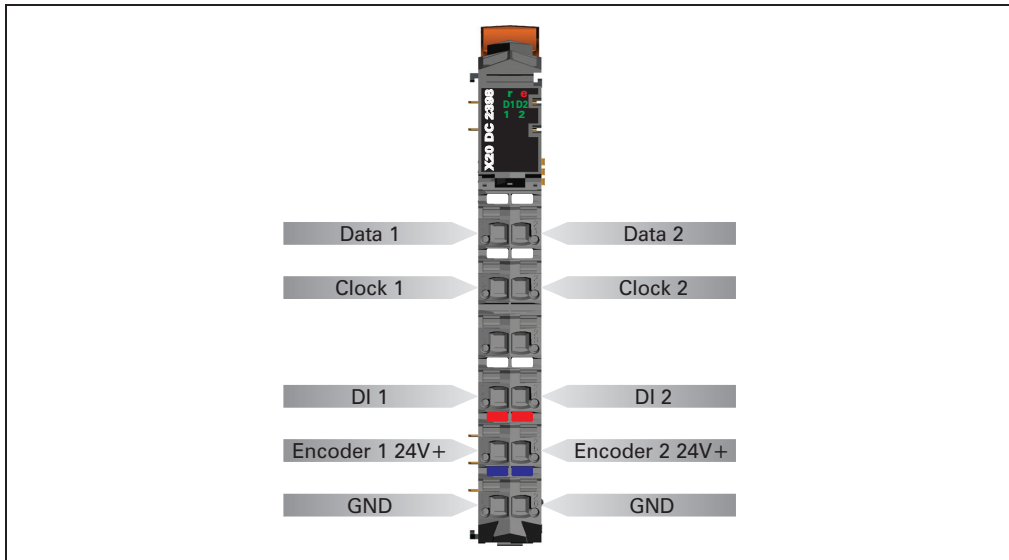


Figure 170: DC2398 - pin assignments

13.7.7 Connection example

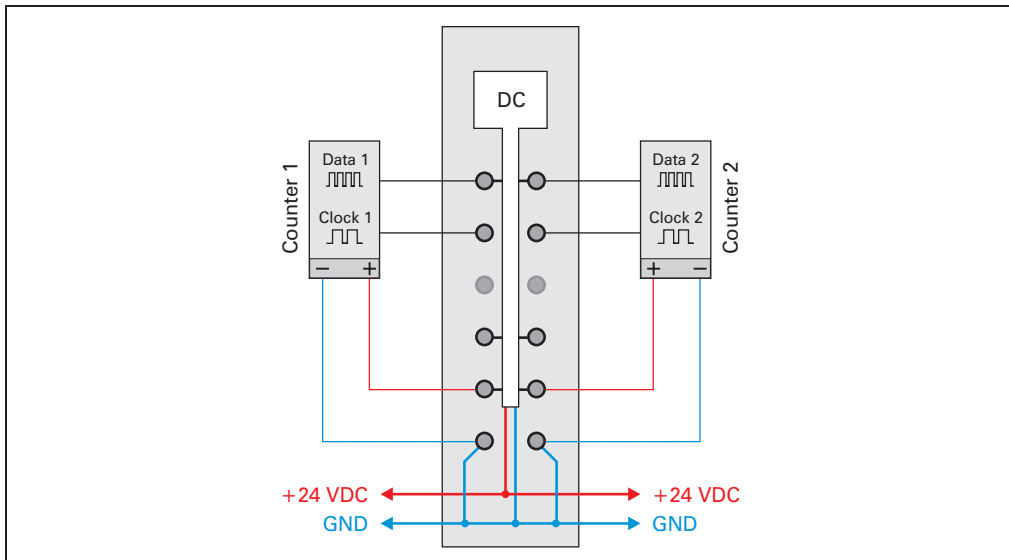


Figure 171: DC2398 - connection example

13.7.8 Input circuit diagram

Counter inputs

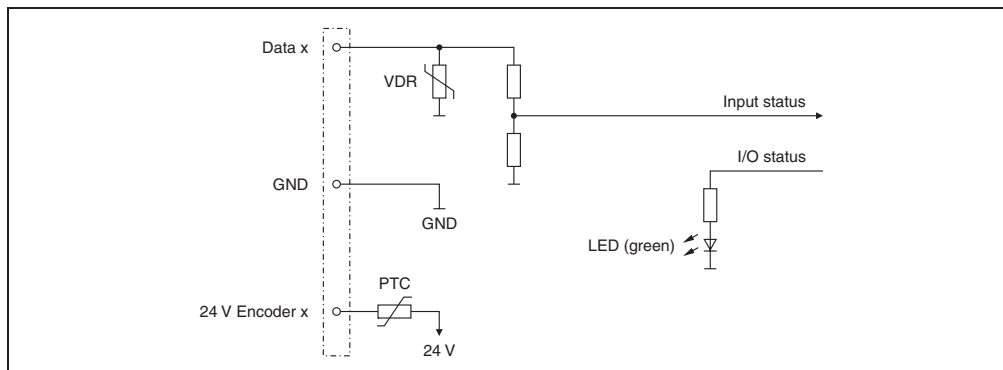


Figure 172: DC2398 - input circuit diagram - counter inputs

Standard inputs

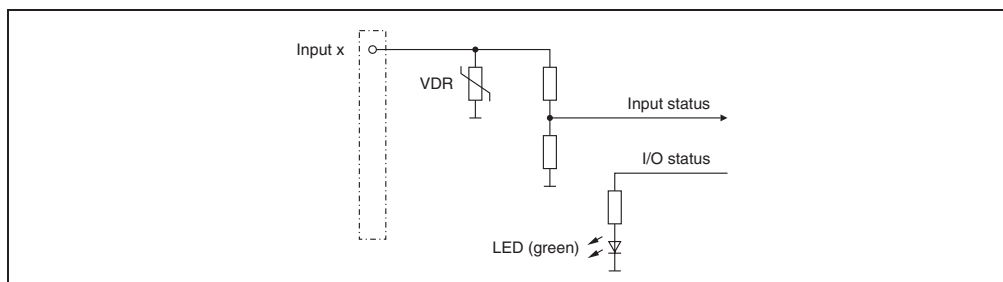


Figure 173: DC2398 - input circuit diagram - standard inputs

13.7.9 Output circuit diagram

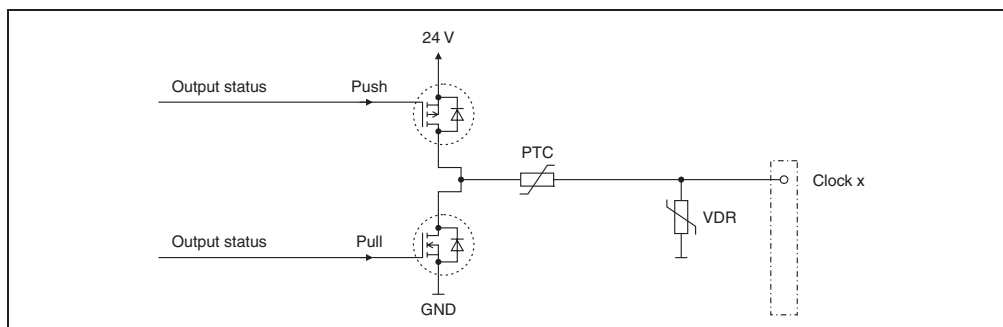


Figure 174: DC2398 - output circuit diagram

13.8 DC2395

13.8.1 General information

The DC2395 module is a multifunctional counter module. It can be connected to an SSI encoder, an ABR encoder, two AB encoders, or four event counters. Two outputs are available for pulse width modulation. The functions can also be mixed (see section 13.8.8 "Connection possibilities" on page 401).

- 24 VDC encoder inputs
- SSI, ABR, AB or event counters for inputs
- Pulse width modulation for outputs
- 24 VDC and GND for encoder supply

13.8.2 Order data

Model number	Short description	Image
	Counter module	
X20DC2395	X20 digital counter module, 1 x SSI absolute encoders 24 V, 1 x ABR incremental encoder 24 V, 2 x AB incremental encoder 24 V, 4 x event counter, or 2 x PWM	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 219: DC2395 - order data

13.8.3 Technical data

Product ID	DC2395
Short description	
I/O module	1 SSI absolute encoder 24 V, 1 ABR incremental encoder 24 V, 2 AB incremental encoder 24 V, 4 x event counter, or 2 x pulse width modulation
SSI absolute encoder	
Number	1
Encoder inputs	24 V, asymmetrical
Counter size	32-bit
Maximum transfer rate	125 kBit/s
Encoder supply	Module-internal, max. 600 mA
Incremental encoder	
Number	2
Encoder inputs	24 V, asymmetrical
Counter size	16-bit
Input frequency (max.)	100 kHz
Evaluation	4-fold
Encoder supply	Module-internal, max. 600 mA
Event counter	
Number	4
Rated voltage	24 VDC
Counter size	16-bit
Evaluation	Each edge, cyclic counter
Signal form	Square wave pulse
Digital outputs	
Number	2
Rated voltage	24 VDC
Rated output current	0.1 A
Total current	0.2 A
Output circuit	Sink or source
Output protection	Thermal cutoff for over-current and short circuit, integrated protection for switching inductances.
Pulse width modulation ¹⁾	
Period duration	41.6 µs ... 500 ms
Pulse length	0 to 100 %
Resolution	0.1%
Actuator supply	Module-internal, max. 600 mA

Table 220: DC2395 - technical data

Product ID	DC2395
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error Outputs	Yes, with status LED and software status Yes, with status LED and software status (output status)
Electrical isolation Encoder - Bus Output - Bus Output - Encoder Encoder - Encoder Output - Output	Yes Yes No No No
Power consumption Bus I/O internal	Typ. 0.01 W Typ. 1.4 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 220: DC2395 - technical data (cont.)

1) Dead time when switching between push and pull: Max. 1.5 µs

13.8.4 Additional technical data

Product ID	DC2395
SSI absolute encoder	
Coding	Gray/Binary
CLK: Output current	Max. 100 mA
Overload behavior of the encoder supply	Short-circuit protection, overload protection

Table 221: DC2395 - additional technical data

X20 System modules • Counter modules • DC2395

Product ID	DC2395
Incremental encoder	
Overload behavior of the encoder supply	Short-circuit protection, overload protection
Event counter	
Input frequency	100 kHz
Count frequency	200 kHz
Input filter Hardware Software	$\leq 2 \mu\text{s}$ -
Input current at 24 VDC	Approx. 1.3 mA
Input resistance	18.4 k Ω
Switching threshold Low High	<5 VDC >15 VDC
Isolation voltage betw. channel and bus	500 V _{eff}
Digital outputs	
Type	Push / Pull / Push-Pull
Switching voltage	24 VDC (-15 % / +20 %)
Diagnostics status	Output monitoring
Leakage current when switched off	Max. 25 μA
Residual voltage	<0.9 V @ 0.1 A rated current
Short circuit peak current	<10 A
Switching on after overload or short circuit cutoff	approx. 10 ms (depends on the module temperature)
Switching delay 0 → 1 1 → 0	<2 μs <2 μs
Switching frequency Resistive load Inductive load	Max. 24 kHz See section 13.8.11 "Switching inductive loads" on page 403 (with 90 % duty cycle)
Braking voltage when switching off inductive loads	Switching voltage + 0.6 VDC
Isolation voltage betw. channel and bus	500 V _{eff}
General information	
B&R ID code	\$1CD4
Input voltage	24 VDC (-15 % / +20 %)
Type of signal lines	Shielded cables should be used for all signal lines

Table 221: DC2395 - additional technical data (cont.)

13.8.5 Status LEDs

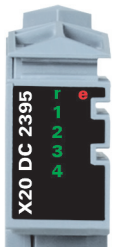
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			On	Error or reset state
	1 - 4	Green		Status of the corresponding digital signal

Table 222: DC2395 - status indicators

13.8.6 Pin assignments

Shielded cables should be used for all signal lines.

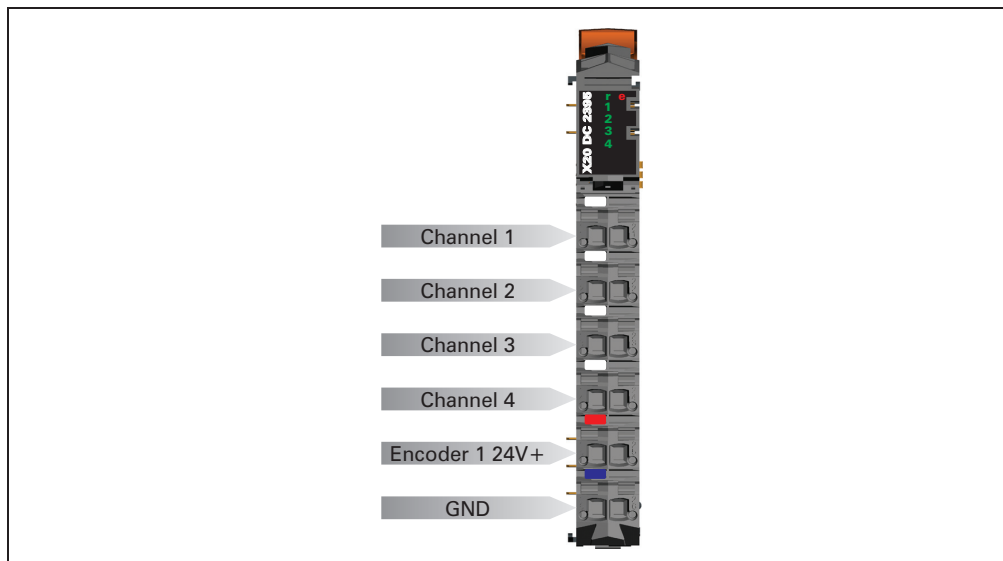


Figure 175: DC2395 - pin assignments

13.8.7 Connection example

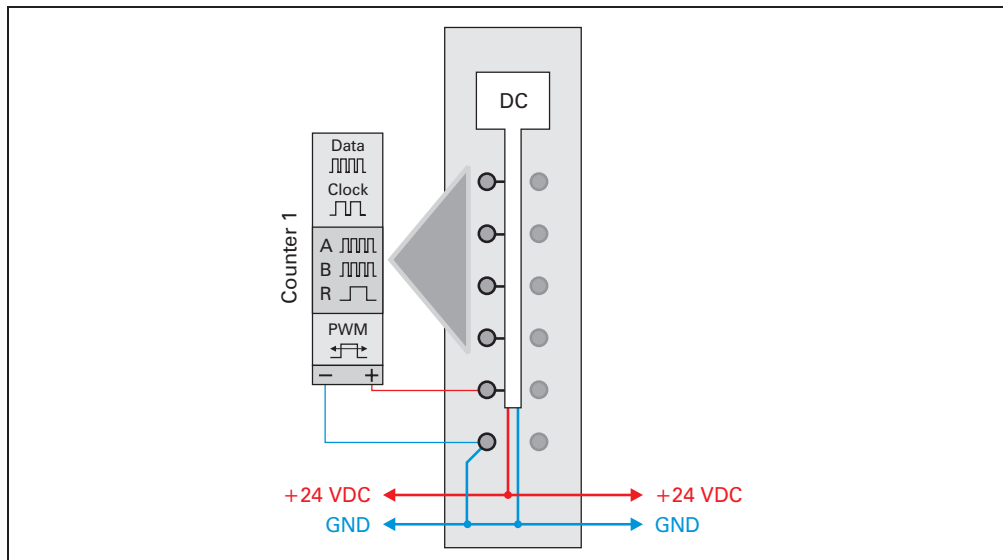


Figure 176: DC2395 - connection example